Fast Simulation-Based Bayesian Estimation of Heterogeneous and Representative Agent Models

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This paper proposes a technique for the estimation of structural macroeconomic models from machine learning: simulation-based maximum likelihood or Bayesian algorithm. This approach, dubbed as sequential neural posterior estimation (SNPE), is able to derive posteriors or the maximum likelihood estimate even when the likelihood function is not tractable. Because the likelihood is not needed for Bayesian estimation, filtering is also not needed. The approach aims to simulate from the joint distribution of parameters and simulated data, then fits a conditional density estimator to get the posterior, $p(\theta|X)$. I show the posteriors estimated on the benchmark Smets and Wouters (2007) are higher quality and faster estimation compared to Metropolis-Hastings. Additionally the ability to perform simulation based maximum likelihood/Bayesian estimation, allows for a more flexible estimation algorithm, that can work with projection and value function iteration. Additionally with HANK models, the algorithm sidesteps lossy dimensionality reduction when using Reiter's method and improves estimation speed of models solved via Winberry method by 13x per CPU iteration. With all this in consideration, SNPE can address the computational expense of Metropolis-Hastings and allows solution methods which don't yield a tractable likelihood to be better estimated. An empirical application estimating a full information HANK model with both aggregate time sereis and cross sectional microdata demonstrates the capability of this technique. A tutorial can be found at: https://colab.research.google.com/drive/14ZDA6fvHlQBF2yJMSS-vXSFSx0QwwZkA?usp=sharing

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I. Introduction

Current Bayesian structural estimation techniques often lack reproducibility for medium to large state-of-the-art structural models. Research has pointed out the reproducibility problem for structural macro models estimated in with the standard Metropolis-Hastings MCMC (MH-MCMC) (Tovar, 2009). In other fields, it is widely known that Metropolis-Hastings or even improvements like Hamiltonian Monte Carlo converge slower that machine learning alternatives (**?**). This convergence rate is very often exponential in rate (Chewi, 2023), which means even when running for 10s of millions of iterations, convergence does not occur. Because MCMC samplers get stuck in local minima without moving for many iterations, this can trick econometricians into thinking the method has converged.

The other downside of MCMC and maximum likelihood based methods is the inability to estimate models with intractable likelihoods with without measurement error, compromising the true model. Dynamic models solved via projection and value function iteration, are models with intractable likelihoods. However in order to estimate models like these, techniques like method of simulated moments (MSM) (McFadden, 1989), estimate models from simulation only. MSM lacks efficiency, estimation is dependent on the econometricians choice of moments to match, and econometricians are forced to impose the Markov assumption that the current time step is only dependent on a few previous lags. My approach, SNPE, can be thought of as an extension to (Kaji, Manresa and Pouliot, 2020), that allows for Bayesian and efficient MLE estimation from simulation, and removes the Markov assumption, while maintaining the efficient estimation via simulation. In the same way, in builds upon SMC Bayesian estimation (Herbst and Schorfheide, 2014) by improving estimation speed/accuracy, ease of use, and the ability to MLE and Bayesian estimate models via simulation only.

As an illustration of its performance, below is the posterior from SNPE on the 36 parameter Smets and Wouters (2007) model. Instead of the Smets-Wouters priors, I've used uniform priors over the same range–making this problem more difficult from an estimation perspective. The red bar indicates the true parameter value, generating the simulated data. I've presented four parameters

that illustrate the different kinds of fit that the model produces. The entire set of posteriors is in the results section.

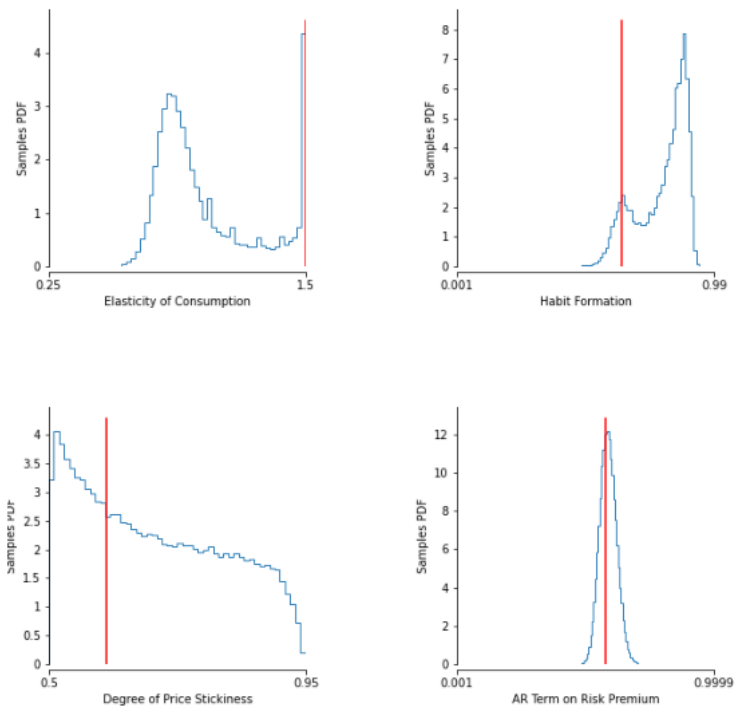

Figure I: Example of SBI posteriorsl

These posteriors were derived from 500 thousand SNPE iterations. The elasticity of consumption plot shows a multi-modal distribution where the mode is exactly at the true parameter value. Likewise the habit formation plot is also multi-modal. The degree of price stickiness is another posterior, however the true parameter value is not at the mode. This will happen because the simulated data was only 200 time steps large to match the identification problems of real macro data. The forth chart indicates the situation where the distribution concentrates around the true parameter value which is the "best" result one wants as a modeler. Unlike MCMC, SNPE naturally generates smooth posterior distributions–the posterior plots are unsmoothed histograms. Additionally, it looks like SNPE can converge to both multi-modal and uni-modal distributions. As a comparison, here is MH-MCMC after 10 million iterations (20x more steps):

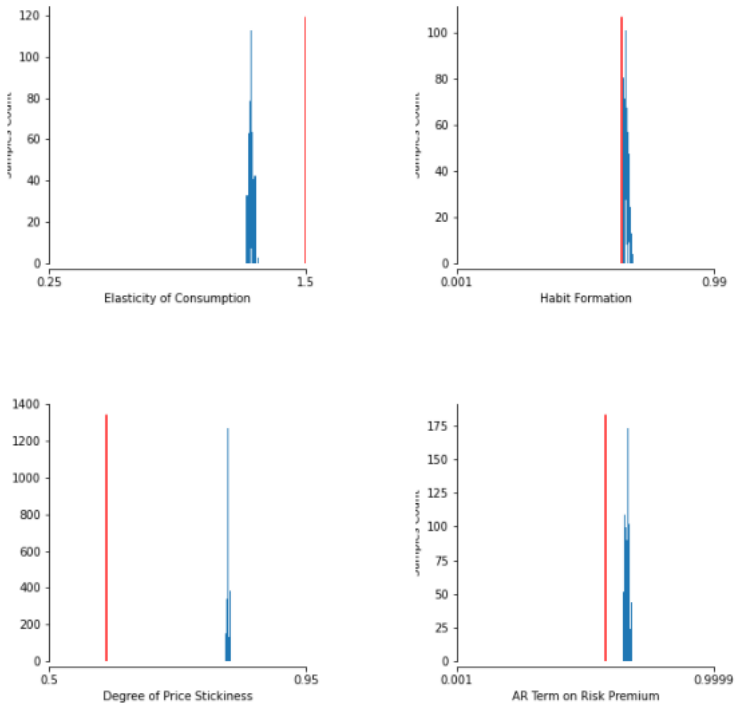

Figure II: Example of MCMC posteriorsl

In the illustration, the MCMC does poor job of exploring the posterior. It seems like it finds a location of relatively high likelihood, but then gets stuck in that area. In three of the four charts, the support of the distribution does not contain the true parameter value. This also suggest even after 10 million recorded iterations over 8 chains and 10000 burn in per chain, that MCMC still hasn't converged and is not close to converging. In particular, MCMC has problems with multi-modality. Theoretically, Chewi (2023) points out that there are no polynomial guarantees for MCMC style sampling when the posteriors are multi-modal and MCMC probably converges at an exponential rate. However, SNPE uses a density estimator to model the posteriors, and as a maximum likelihood density estimator, it could be reasonably expected to converge at the parametric rate $O(n^{-1/2})$, although given some regularity conditions it might be somewhat slower but still likely polynomial (Farrell, Liang and Misra, 2018). Empirically, while the posterior shown may not quite be state of the art for MCMC Bayesian structural estimation, estimation accuracy in Bayesian economics

with MCMC more often resembles the poor MCMC distribution other than for small toy problems. Although they may be out there, I have never seen posteriors that look as accurate as the SNPE approach.

Another advantage over current methods is the flexibility the algorithm has to handle any solution method. Since the method learns the likelihood/posterior from simulation and doesn't require a likelihood function, as long as a solution method generates a model that can be simulated from (all models can be simulated from to the best of my knowledge) the approach can be Bayesian or MLE estimated. The below chart illustrates the breadth and depth of the models this algorithm can Bayesian and MLE estimate:

Solution Method/Algorithm	**Details**	**SNPE**	**MCMC**	**MSM**
Vanilla Perturbation	Posterior Quality	✓	∼	✗
	Ability to Estimate	✓	✓	∼
Vanilla Projection	Posterior Quality	✓	∼	✗
	Ability to Estimate	✓	∼	∼
Value Function Iteration	Posterior Quality	✓	∼	✗
	Ability to Estimate	∼	∼	∼
HANK Reiter	Posterior Quality	✓	∼	✗
	Ability to Estimate	✓	∼/✗	∼
HANK Winberry	Posterior Quality	✓	∼	✗
	Ability to Estimate	✓	∼	∼

Here is the legend:

Symbol	Meaning
✓	Model can do this without difficulties
∼	Model can do inaccurately or with some modifications
✗	Model not designed to do this

The chart compares three approaches. SNPE, MCMC, and MSM. A chart also illustrating sequential Monte Carlo estimation (not as an intermediate step in finding the state space likelihood but used for posterior estimation) is in the appendix. MSM is a frequentest approach and doesn't

generate posteriors, which is why there are x marks in those cells. It also can estimate all solution methods via simulation, but it isn't efficient and imposes the Markov assumption on the data. MCMC has poor posterior quality as mentioned above, but can estimate small to midsize models solved via perturbation fairly well. However because it requires a likelihood function it cannot estimate other models without something like measurement error–fundementally changing the model. Additionally its ability to estimate models solved via the Reiter's method (Reiter, 2009), require the use of dimensionality reduction (Ahn et al., 2018), which loses information. SNPE can estimate perturbation and projection methods without measurement error. Likewise it should theoretically be able to do the same with value function iteration, but I haven't yet been successful. As the method requires no likelihood function and only simulated data, the method doesn't require dimensionality reduction when estimating a model solved via Reiter's method. It can Bayesian estimate Winberry method 13x faster than state of the art (Liu and Plagborg-Møller, 2023) and ultimately extends the Liu and Plagborg-Møller (2023) paper to an empirical application.

Following Kristensen and Shin (2012), the method begins by simulating from the prior, $P(\theta)$ and then drawing simulated data, X', from the model conditional on the prior to generate samples from the joint distribution, $X', \theta \sim P(X', \theta) = P(X'|\theta)P(\theta)$. A machine learning density estimator, such as a conditional mixture of Gaussians (Bishop, 1994), a normalizing flow (Rezende and Mohamed, 2015), or Generalized Adversarial Networks (GAN) (Goodfellow et al., 2014), is then used to estimate the conditional density $P(\theta|X')$ from the joint samples $(X', \theta) \sim P(X', \theta)$. A description of mixture of Gaussian and a normalizing flow will be discussed later on. The GAN approach is discussed in the appendix. The density estimator is then conditioned on the observed X data, giving an estimate of the posterior: $P(\theta|X' = X)$.

The general approach around simulation-based inference has gained popularity in many fields outside of structural estimation. The approach is often used for Bayesian inference in machine learning (Durkan, Murray and Papamakarios, 2020), (Greenberg, Nonnenmacher and Macke, 2019). The approach has also been used in neuroscience (Boelts et al., 2021) and ecology (DiNapoli et al., 2021). Furthermore, the technique has also become one of the leading estimation methods in many fields of physics (Brehmer, 2021), (Cranmer, Brehmer and Louppe, 2020). Economics could

benefit from adopting such a widely proven technique. In the case of large models like Smets and Wouters (2007) or time consuming HANK models, my approach yield more accurate and reproducible posteriors, faster than traditional MH-MCMC, and more flexibility estimate models using any choice of solution technique.

II. Literature Review

The aim of this literature review is to provide a picture of where my paper fits in economics. The first topic is simulation-based methods in economics, including their history, development, and applications. Then I will delve into the solution and estimation of heterogeneous agent models, including their relevance and current state of the literature. I will cover the concepts of MH-MCMC and machine learning and their applications in the section following this literature review.

II.A. Simulation-Based Models in Economics

Simulation-based estimation of models, structural or otherwise, has been a prominent approach in economics for many years. Originating from MSM (McFadden, 1989) or Generalized Method of Moments, GMM (Hansen, 1982), simulation-based likelihood and inference techniques have been developed (Pakes and Pollard, 1989), (Duffie and Singleton, 1990) and improved to address the difficulty of model estimation. As a simulation techique, SNPE is more closely builds upon MSM. However, GMM focuses on matching moments, but MSM extends GMM by using simulation from the model to match moments. Maximum simulated likelihood (Lerman and Manski, 1981) (Hajivassiliou et al., 1997), (Haan and Uhlendorff, 2006) and other techniques have been used to improve efficiency, but still require the use of measurement error. Researchers have adopted these simulation-based methods to estimate dynamic models, particularly in industrial organization (Keane and Wolpin, 1994), as well as the ever-present usage for structural models in macroeconomics (Herbst and Schorfheide, 2015) and finance (Hennessy and Whited, 2007).

Providing a estimation routine that can handle many different types of solution methods is a challenge in economics. Three widely utilized methods for solving these models are perturbation,

projection, and value function iteration, each with its own strengths and limitations (Fernández-Villaverde, Rubio-Ramírez and Schorfheide, 2016), (Judd et al., 2017). Perturbation is well-suited for models where the policy function is smooth. One benefit is that this solution approach yields likelihood functions that allow for Bayesian and MLE estimation using conventional techniques. However, for highly nonlinear models with large shocks or non-analytic functions, projection and value function iteration are preferred (Terry, 2017). Unfortunately, for a significant number of models including those solved via value function iteration (Hennessy and Whited, 2007), heterogeneous agent models with intractable likelihoods (Kukackaa and Barunika, n.d.), and most projection models, the likelihood is intractable and practitioners must rely on less statistically efficient point estimate methods that are subject to limitations compared to Bayesian and MLE estimation. The likelihood intractability is often due to algorithmicly transforming normal shocks with nonlinear operations. The estimation of non-perturbation models without measurement error typically requires MSM and related approaches, which cannot perform Bayesian and in many case even full-information frequentist estimation.

The issues of MSM have pushed the literature to search for ways to improve robustness to misspecification and efficiency. Even if a model is misspecified, one would like to obtain as narrow standard error bars as possible. Efforts have been made to develop efficient method of moment estimators using a spectrum of moments to guarantee that data is in the support of distributions (Gallant and Tauchen, 1996). An additional approach that this paper builds upon is Kaji, Manresa and Pouliot (2020), which allows for efficient estimation of structural. All these approaches do not have Bayesian extensions. However, the more pernicious problem is that all these estimation approaches assume that only a small number of lags is a sufficient statistic for the entire history of the data. Thus one estimates this time series as a panel problem. One appends a small number of lags to the time step in question, then feed every time-step into the model, as one would when estimating a VAR. I will call this assumption the Markov assumption. Importantly, my approach does not impose this, data from as far back as one wants, can provide information for estimation. A structural model naturally can be modeled as a state space where time steps as far back in the past as one wants can effect estimation. Thus my approach estimates a model in a way that maintains

faithfulness to the assumptions of a structural model. i will call this assumption the state space assumption.

There are two research agendas that this paper seeks to address. The first is how to build simulation based inference models, that are efficient, can be extended in Bayesian estimation, and avoid the simplifying Markov assumption. The second aspect is to improve the speed/accuracy trade of of MCMC and other Bayesian estimation. This paper aims to address both threads by proposing a method that can provide the advantages of simulation with the efficiency and also provide the ability to Bayesian estimate models faster and more accurately than current methods. Finally this method is even more effective when used in state-of-the-art heterogeneous agent models.

II.B. Heterogeneous Agent Models

The literature surrounding full information estimation of heterogeneous agent models is growing and an essential part of the future of macroeconomics. This section will focus on improving speed and accuracy of estimation methods. The demand for more accurate estimation approaches is significant, given the complexity of the models beings solved. For example, Ahn et al. (2018) proposes to use a PCA based approach to reduce the dimensionality of the latent space. This is necessary because the Kalman filter rapidly becomes intractable as the state space grows. SNPE doesn't need to calculate the likelihood, can skip the Kalman or particle filter, and thus can be estimated without a dimesionality approach that throws away information.

Another aspect of the heterougenous models is the emphasis on computational speed of estimation. There are many papers proposing better solution methods (Auclert et al., 2021*a*), (Winberry, 2018), (Khan and Thomas, 2008), but their complexity means that it is more difficult to estimate with conventional methods. One innovations on this front is Liu and Plagborg-Møller (2023) which uses full information microdata to complement macro variable estimation to perform true full-information Bayesian estimation. In particular they estimate a Krusell and Smith (1998) model via Winberry (2018) in a Bayesian manner. It is difficult to estimate more than the three parameters (β, σ_{meas}, and μ). (Liu and Plagborg-Møller, 2023) don't seem to, and I am unable to as well. Parra-Alvarez, Posch and Wang (2020) highlight similar issues with other heterogenous agent mod-

els. However, on those three parameters, my approach is 13x faster per CPU iteration and converges equally well if not better after 10000 iterations. The issues highlighted by estimation–lack of accuracy, compuational burden, and indentification issues–are reasons so many heterogeneous agent modelers resort to calibration (Auclert et al., 2021*b*), (Ottonello and Winberry, 2020), (Caglio, Darst and Kalemli-Özcan, 2021). There is a need among both heterogeneous agent and representative agent model practitioners to reduce the barriers to maximum likelihood estimation and Bayesian estimation, and the SNPE and related algorithms may help address these barriers.

III. Background on SNPE and MH-MCMC

In recent years, Bayesian inference has gained increased prominence as a method for estimating models in economics. This section sheds light on two prominent techniques in this field, namely SNPE and the economic benchmark MH-MCMC. To provide a comprehensive understanding, a brief overview of Bayesian estimation is first presented. The SNPE estimator, which relies on simulating data from the model being estimated and calculating the posterior using machine learning density estimators, is then discussed.

Despite its success in various applications, the last section will discuss the limitations and pathological cases for the SNPE estimator. While these methods have been widely used, it is essential to understand their limitations and make informed decisions when choosing the most appropriate approach for a particular problem, considering the assumptions and limitations of each method.

III.A. Bayesian Basics

In this subsection, I will discuss the general paradigm of Bayesian estimation, then I will discuss MH-MCMC.

Bayesian estimation attempts to find the posterior given the likelihood and prior of a model using Bayes' rule:

$$P(\theta|X) = \frac{P(X|\theta)P(\theta)}{P(X)}$$

Here $P(\theta|X)$ is the posterior, $P(X|\theta)$ is the likelihood, and $P(\theta)$ is the prior. All approaches that perform Bayesian inference, ranging from MH-MCMC, variational inference, to simulation-based inference concern techniques for calculating $P(\theta|X)$ without calculating the partition function, $P(X)$.

MH-MCMC is a commonly employed technique for Bayesian inference in economics, as described in (Herbst and Schorfheide, 2015). The way MH-MCMC gets around the intractable partition function is to evaluate the likelihood ratios between different parameter values, allowing for the determination of the relative likelihood of different points in the parameter space. As this is a likelihood ratio, one doesn't need to calculate the normalizing partition function, $P(X)$. This information is then utilized to direct a random walk, with points being visited in proportion to the relative likelihoods, effectively sampling from the posterior distribution $P(\theta|X)$. The benchmark method of Bayesian inference in economics is the use of this MH-MCMC algorithm (Herbst and Schorfheide, 2015).

III.B. Sequential Neural Posterior Estimation (SNPE)

This paragraph explains the details of the SNPE algorithm. The algorithm commences by drawing a sample, θ, from the prior distribution, $P(\theta)$. Subsequently, simulated data, X', is generated from the model likelihood, $P(X'|\theta)$. The concatenation of these two simulations results in samples from the joint distribution, $X', \theta \sim P(X', \theta) = P(X'|\theta)P(\theta)$. These samples can then be used to estimate the posterior distribution, $P(\theta|X')$, through the utilization of either a conditional density estimator like a mixture of Gaussians (Bishop, 1994), a normalizing flow density estimator (Mohammad-Djafari and Ayasso, 2009) or a GAN (Goodfellow et al., 2014). The coditioning the conditional density estimator $P(\theta|X')$ on the real data, returns an estimate for the posterior: $P(\theta|X' = X)$. The mixture of Gaussian will be discussed in the following section while the normalizing flow and GAN approach are discussed in the appendix.

Instead of sampling from the prior, multi-round inference will reduce the variance on the estimator. Instead of sampling from the prior, θ is sampled from the best approximation of the posterior, leading to a more accurate density estimator. However, this process requires importance sampling

to correct for the shift in distribution as θ is no longer being drawn from the prior. The first round entails sampling from the prior, while later rounds involve sampling from the updated estimate of the posterior and applying importance sampling to account for the distribution shift. This can be mathematically expressed as $p'(\theta|X') = p(\theta|X')\frac{p'(\theta)}{p(\theta)}$, where $p'(\theta)$ is the proposed prior, $p(\theta)$ is the true prior, and $p'(\theta|X')$ is the estimated posterior obtained by sampling from the proposal distribution. The use of a flexible machine learning density estimator, such as a normalizing flow or a conditional mixture of Gaussians, allows for the unbiased estimation of the posterior through the estimation of $f_\phi(\theta|X')$ in place of $p(\theta|X')$ in the above equation. Papamakarios and Murray (2016) and Greenberg, Nonnenmacher and Macke (2019) have demonstrated that the use of universal approximators, such as normalizing flows (Huang et al., 2018) and conditional mixture of Gaussians (Bishop, 1994), as density estimators in multi-round SNPE allows for the approach to converge towards the true Bayesian posterior in the computational limit under the KL divergence. Below is a description of the SNPE algorithm:

Algorithm 1: SNPE Algorithm

Input: Simulator $p(x|\theta)$, prior $p(\theta)$, data x_0, flow $f_\phi(x|\theta)$, Rounds R, Samples S;

Initialize: Posterior $p^{(0)} = p(\theta)$, data set D = {};

for $i \leftarrow 1$ **to** R **do**

  Sample $\theta^{(n)} \sim p^{(i-1)}$ for $n = 1...S$ with Monte Carlo;

  Simulate $X'^{(n)} \sim P(x|\theta^{(n)})$ for $n = 1...S$;

  Concatenate data $D = D \cup \{X'^{(n)}, \theta^{(n)}\}_{n=1}^{S}$;

  while $d_\phi(X', \theta)$ *not converged* **do**

    Sample $\{X'^{(i)}, \theta^{(i)}\}_i^B \sim D$ from D;

    Train $f_\phi(\theta|X')\frac{p^{(i-1)}}{p(\theta)}$ on $\{X'^{(i)}, \theta^{(i)}\}_i^B$;

  end

  Update posterior $p^{(i)} \propto f_\phi(\theta|X' = X)$;

end

III.C. Density Estimation

The SNPE approach proposed in this article utilize three distinct density estimators: Normalizing Flows, Conditional Mixture of Gaussians, and GANs. These methods are capable of handling high dimensional estimation in X' and θ. The structural models are used to generate samples from X' and θ, then with the exception of GANs, which uses the slightly different SNVI algorithm, each density estimator serves as the $f_\phi(\theta|X')$ function in the above algorithm.

One of the primary benefits of the proposed estimation algorithm is the improved handling of high dimensional estimation of θ. This allows for Bayesian and MLE estimation of models with up to 36 parameters, such as the Smets and Wouters (2007) model. This is a significant improvement over previous works, such as Kristensen and Shin (2012), which are limited in their ability to handle high dimensional estimation. Additionally, this algorithm improves over Kaji, Manresa and Pouliot (2020), which uses a GAN for likelihood evaluation, compared to this approach which uses the GAN as estimator for the joint/posterior.

Each of the three density estimator approaches has its own strengths and weaknesses. Normalizing flows can handle exotic distributions with almost arbitrary dimensionality in θ and X. On the other hand, mixture of Gaussians is the simplest model to implement and can handle most lower dimensionality Bayesian posteriors with smooth and multi-modal distributions. The GAN approach combined with variational inference is designed specifically for structural models that fail to converge regularly within the parameter bounds. Unlike the other two methods, which fit a density estimator directly on the data, the GAN approach aims to mimic the real data, thus placing zero probability mass in areas in the parameter set that lack training data, due to for example non-convergence of the solution method. The other methods will extrapolate pdf values in areas without training data (but in the training data range), potentially behaving strangely in these areas. This results in a slower and less accurate method, but with the benefit of disciplining distributions in areas with no training data.

III.D. Conditional Mixture of Gaussians

The conditional mixture of Gaussians approach is a powerful tool for density estimation in high dimensional spaces and, of the three methods, is the easiest to explain. As outlined in Bishop (1994), a mixture of Gaussians is a linear combination of N Gaussian distributions, each with its own mean, covariance, and weight. The probability density function (pdf) of a point under this mixture can be calculated using the equation:

$$P(\theta) = \sum_i \pi_i * Q_i(\theta)$$

Where $Q_i(\theta)$ is a normal distribution with mean and variance (covariance) specific to each Gaussian component. This represents the unconditional density estimator, however, the algorithm requires a conditional density estimator. This can be achieved by using an optimizable function, such as a neural network, that takes the X' data as input and returns the parameters for the mixture of Gaussians. The network would return the weights $\pi_i(X')$, means $\mu(X')$, and covariances $\sigma(X')$ for each Gaussian component, resulting in the following conditional density estimator:

$$P(\theta|X') = \sum_i \pi_i(X') * Q_i(\theta|\mu(X'), \sigma(X'))$$

Finally here is an estimation of conditional density estimators. The parameters of the neural network can be optimized via maximum likelihood estimation (MLE), maximizing $P(\theta|X')$ and improving the accuracy of the density estimator. The approach maximizes $P(\theta|X')$ given the simulated conditioning data X'. The X' is inserted into the model, a feedforward neural network will learn a mapping from X' to the parameters of the mixture of Guassinas (ie $\sigma(X')$, $\mu(X')$ etc.). The parameters of the feedforward neural network is then optimized so that the mixture of Guassians produces the highest likelihood for the θ variables. The details of the normalizing flow and GAN approaches are described in the appendix.

III.E. Caveats and Pathological Examples Using SNPE

There are certain limitations associated with SNPE that must be considered when applying it to a specific problem. One of the limitations of SNPE is its inability to handle the discontinuities that are present in some models with kinks, such as s-S models. This leads to a lack of smoothness in the posterior of the kinked models, and the assumption that there is a normalizing flow parameterization that can generate the posterior is violated. In practice, a flow still converges, but it learns a continuous function that approximates the discontinuity in the posterior. This approximation can still be near the true likelihood, but if estimation with a discontinuous model is a problem, one can also add measurement error to smooth the problem out.

Another limitation of SNPE is its difficulty in handling stochastic singularities, which also result in estimation issues. When this occurs, the dynamic model learns a lower dimensional sub-manifold in the entire space, and the normalizing flow or mixture of Gaussians has to have positive support on the entire space of the target distribution. This also violates the assumption that the machine learning density estimators can generate the posterior. In cases where there are stochastic singularities in a model estimated on real data, the model will fail to match the data. However, one can add more shocks, add measurement error, or make a change in modeling to avoid a stochastic singularity (Caballero and Engel, 2007) and allow full support over the range of the model output.

IV. Results

This section will evaluate the performance of SNPE by examining the proper behavior of the posterior and inspecting its accuracy. To this end, seven models are estimated using the SNPE algorithm, each showcasing a unique aspect and characteristic of the algorithm. The first model is the RBC model that is estimated on both simulated and real data. The second model is a toy corporate finance model that is solved via value function iteration. The third model is the Lucas asset pricing model solved through projection. The fourth model is the Smets-Wouters model, which is estimated through both the MH-MCMC and SNPE algorithms. The fifth model is the HANK model solved through Reiter's method and time iteration. The sixth model is a bequests model

solved via value function iteration. The seventh model improves on full information estimation of a Winberry (2018) model speeding up the approach taken by Liu and Plagborg-Møller (2023). The details of the construction and implementation of these models can be found in the appendix.

In the empirical analysis, the accuracy of the posterior is inspected by comparing it with the calibrated parameters producing the simulated data. In theory, through the Bernstein-von Mises theorem, the posterior should converge to the true values, although perhaps the structural models may not satisfy the assumptions of the theorem. Although the correct posterior should converge to the true parameter value as the number of data samples increases, finite sample sizes may not always result in a perfect match. Hence, it is important to assess the posterior mode and ensure that it generally matches the true value and that there is no vanishing posterior mass at the true values. This check provides useful information to determine the performance of SNPE in practice and how it compares with other estimation techniques, such as the MH-MCMC algorithm. While in many cases the method is tested using simulated data, in order to resemble real world data, 200 time steps are sampled from the simulated model, to maintain a rough correspondence to the length of real world macroeconomic data. This small number of time steps implies that some parameters will be inadequately identified especially in the models with a large number of parameters. For the sake of illustrating the estimation benefits, all priors will be uniform over the range specified in the graph, making the estimation problem difficult to demonstrate modeling capability.

IV.A. RBC Model

In the present article, the results of estimating the RBC model using the SNPE algorithm are presented and discussed. The RBC model is a standard model in macroeconomics and it has four parameters, α, β, δ, and ρ, with a fixed CRRA elasticity of 2. The first 100 iterations are dropped to attain a steady-state behavior. The results of the posterior are displayed below, where the red bar represents the true parameter values and the blue line represents the posterior as estimated by the simulation-based inference approach:

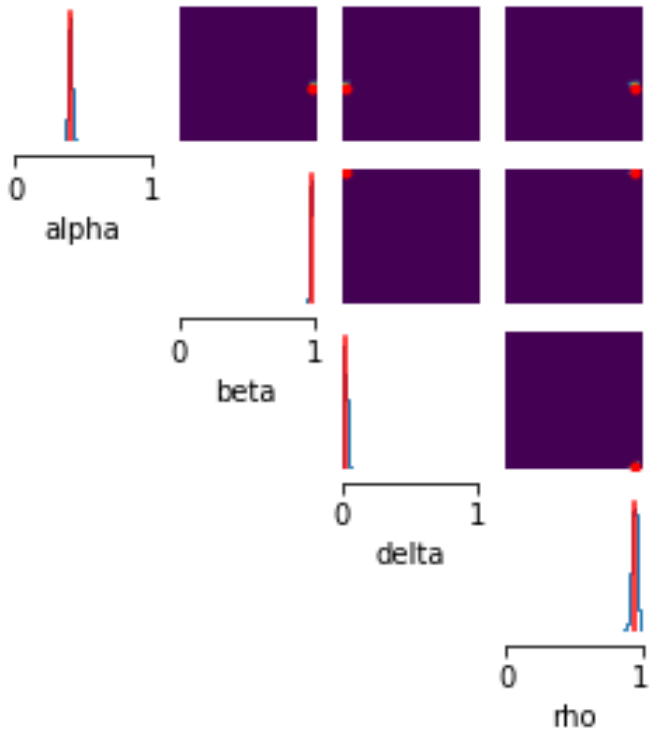

Figure III: Simulated Data RBC

The diagonal density plots in the upper triangular portion of the graph display two-way densities, with the color indicating the probability mass. The approximate ground truth is obtained by running the MH-MCMC for 2 million iterations as show below:

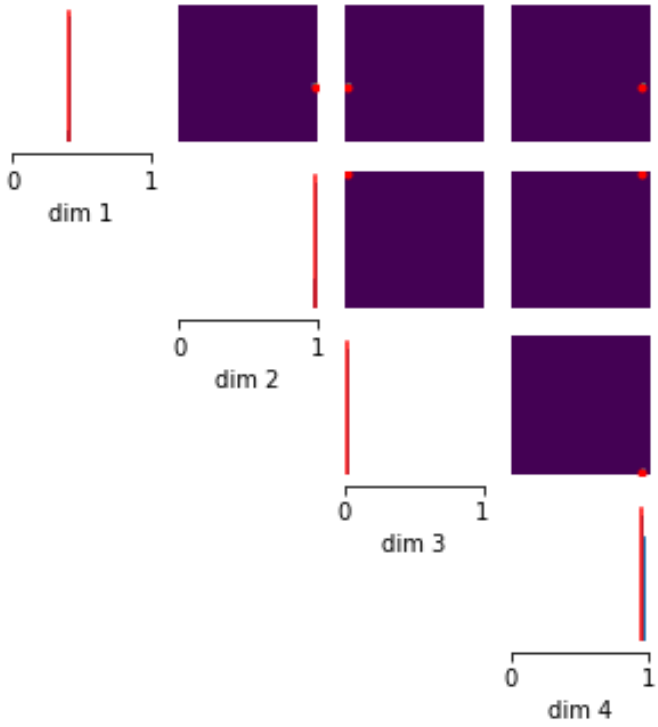

Figure IV: Simulated Data RBC

Below the same RBC model is also estimated on real data, where there is no available ground truth. The parameters derived from the real data are not necessarily realistic, which is expected as the RBC model is a simple model. The method is uncertain about the ground truth, as indicated by the spread of the posterior. The model also predicts some parameters that are not in line with theory, such as the discount rate, β, and the parameter on the productivity process, ρ. The depreciation rate, δ, is the only parameter that is relatively reasonable. Below is the posterior estimated on real data:

In conclusion, the SNPE algorithm is applied to estimate the RBC model and the results are displayed and discussed. The results indicate that the SNPE algorithm can estimate the RBC model to a good degree of accuracy. The model's limitations in real-world applications are also highlighted, as it fails to estimate some parameters that are not in line with theoretical predictions.

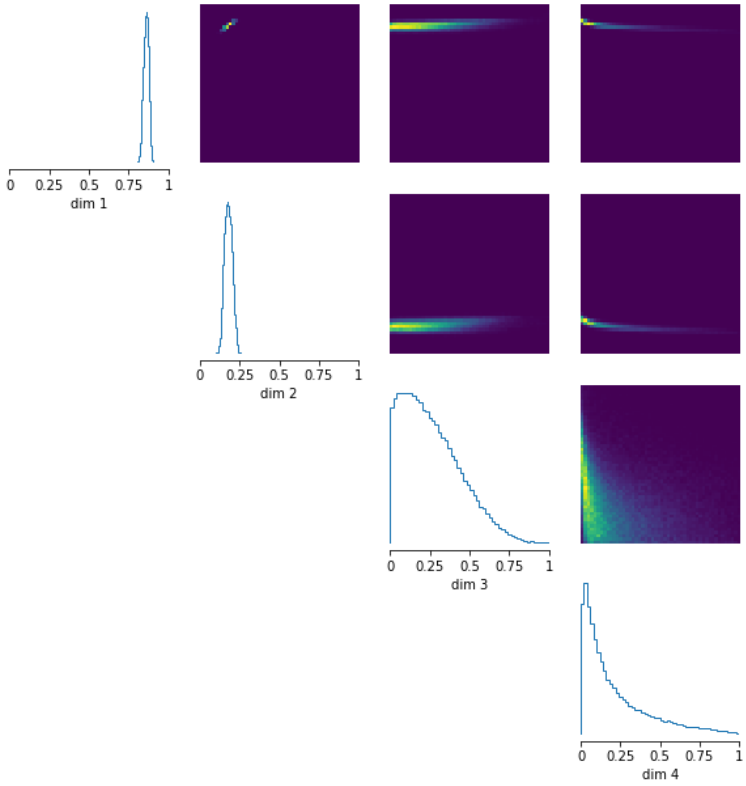

Figure V: Real Data RBC

This highlights the need for more complex models to better reflect real-world economic scenarios, which will be illustrated in the next models.

IV.B. Cash Flow Model

in this section, I am estimating a partial equilibrium model of firm cash flow by utilizing value function iteration. The model is specified with four parameters, α, δ, σ, and ρ. α is the capital share in the production function. δ is depreciation rate of capital. σ is the standard deviation of log productivity AR process. ρ is the slope of the log productivity AR process. The interest rate was set at 5 percent and the discount factor was calculated as 1/(1+r).

To estimate the model parameters, two different estimation methods were applied, including a simulation-based inference approach using SNPE and one approach using a an extension of SNPE but with a GAN. density estimator along with a variational inference sampling scheme. This technique is called sequential neural variational inference (SNVI). From the visual inspection of the charts, it can be observed that the posteriors obtained from the two estimation methods are similar, suggesting that the same distribution was learned. Additionally, the mode of the distribution appears to be close to the actual parameter values, which indicates that the model is capable of accurately recovering the true parameters.

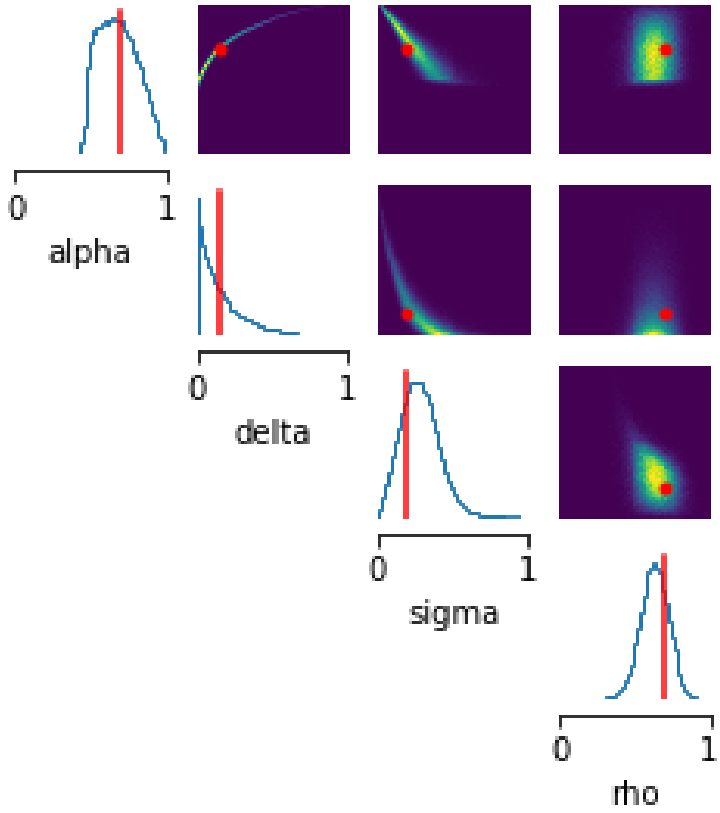

Figure VI: Value Function Iteration with a Normalizing Flow Density Estimator

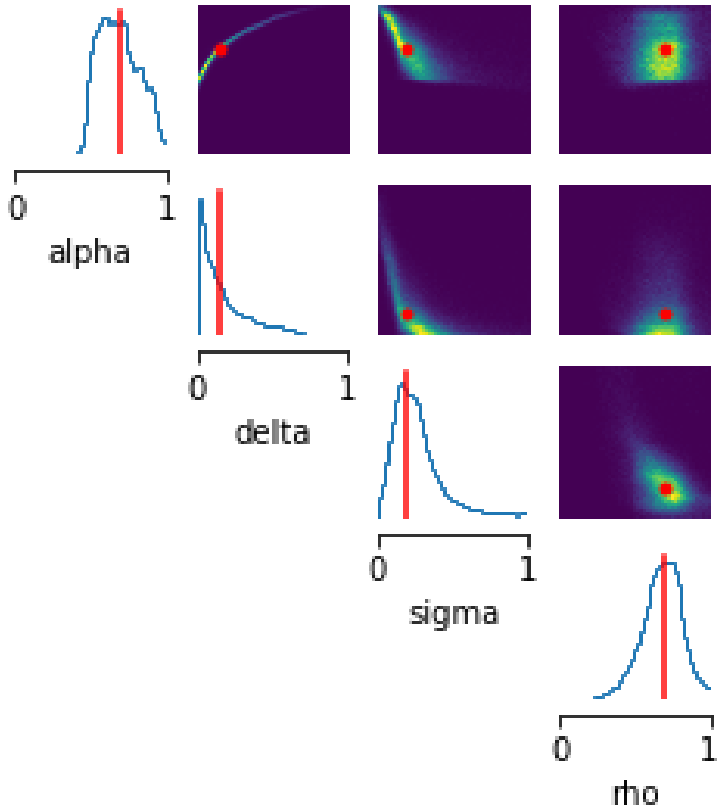

Figure VII: Value Function Iteration with a GAN Density Estimator

The use of the normalizing flow and the GAN density estimator highlights the versatility of this estimation framework and demonstrates the potential for future work in this area. The technical details of SNVI and its simplified version, Sequential Neural Ratio Estimation (SNRE), are discussed in the appendix for further reference.

IV.C. Lucas Asset Pricing Projection Model

In the following section, we demonstrate the estimation procedure on a Lucas asset pricing model, which is solved via projection without measurement error. The purpose of this estimation is to showcase the capability of the SNPE approach to estimate models of this nature:

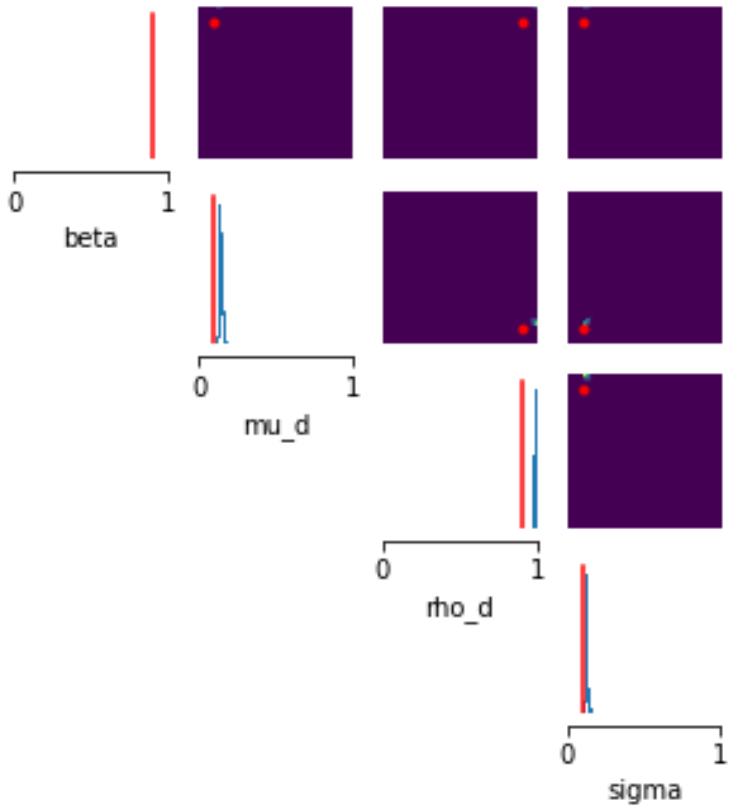

Figure VIII: Ten Parameter HANK model

This estimation procedure took under 10 hours to complete on an 8-core Intel i7 machine, highlighting the computational efficiency of this approach. This result also demonstrates that models solved via projection can be effectively estimated without measurement error using the SNPE framework. The posterior accuracy is pretty good, however it seems like the losteriors have a mode that is slightly shifted and while close to the true parameter value, don't put much support the on thrue value along the μ_d and ρ_d dimensions.

IV.D. Smets-Wouters 2007 Model

The Smets-Wouters (2007) DSGE model is a representative agent New Keynesian model that describes an economy with a flexible price sector and a sector with sticky Calvo prices. The

model has eight observed variables and eight shocks that interact with the rest of the model in an autoregressive manner. The linearized form of the model has 36 parameters, which were estimated using roughly 500,000 samples generated by the posterior. The parameters are separated into three different charts of 12 parameters each, and only the marginals are displayed to avoid visual clutter. These graphs are histograms rather than smoothed with kernel. This illustrates that the density estimators, be they Gaussians or flows, generate natural looking posteriors, which is not something MH-MCMC generates.

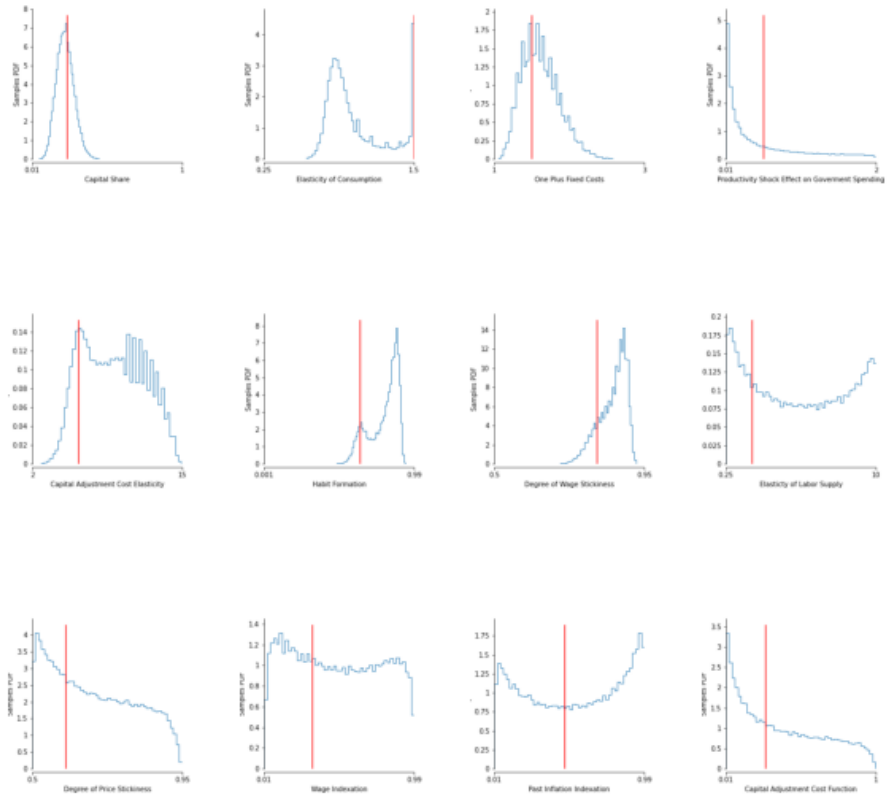

Figure IX: Smets-Wouters Posterior with Parameters 1-12

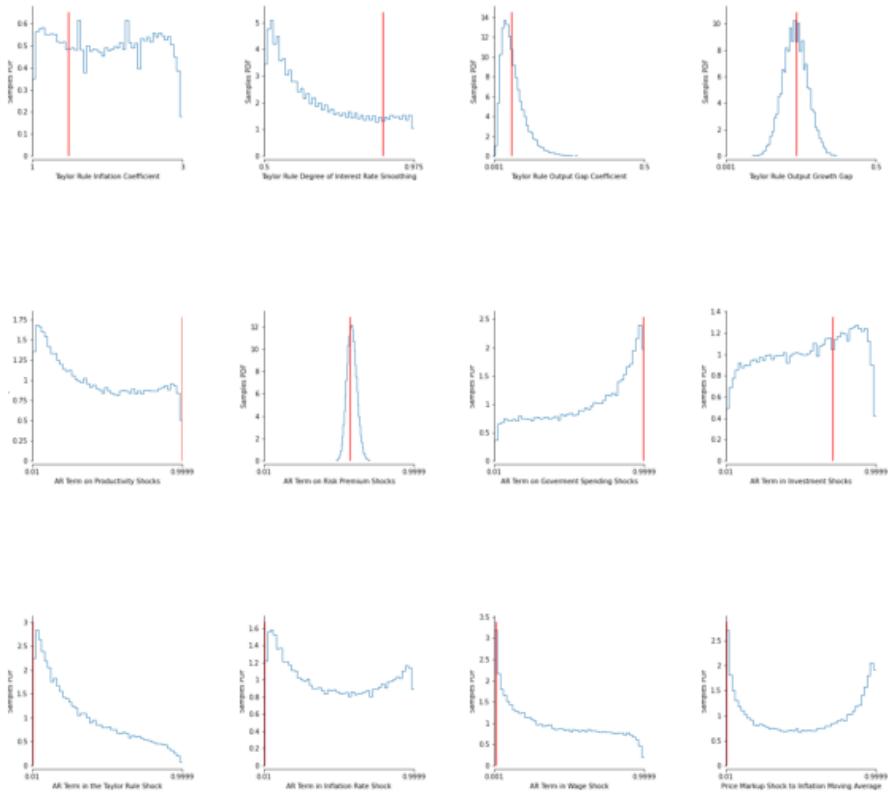

Figure X: Smets-Wouters Posterior with Parameters 13-24

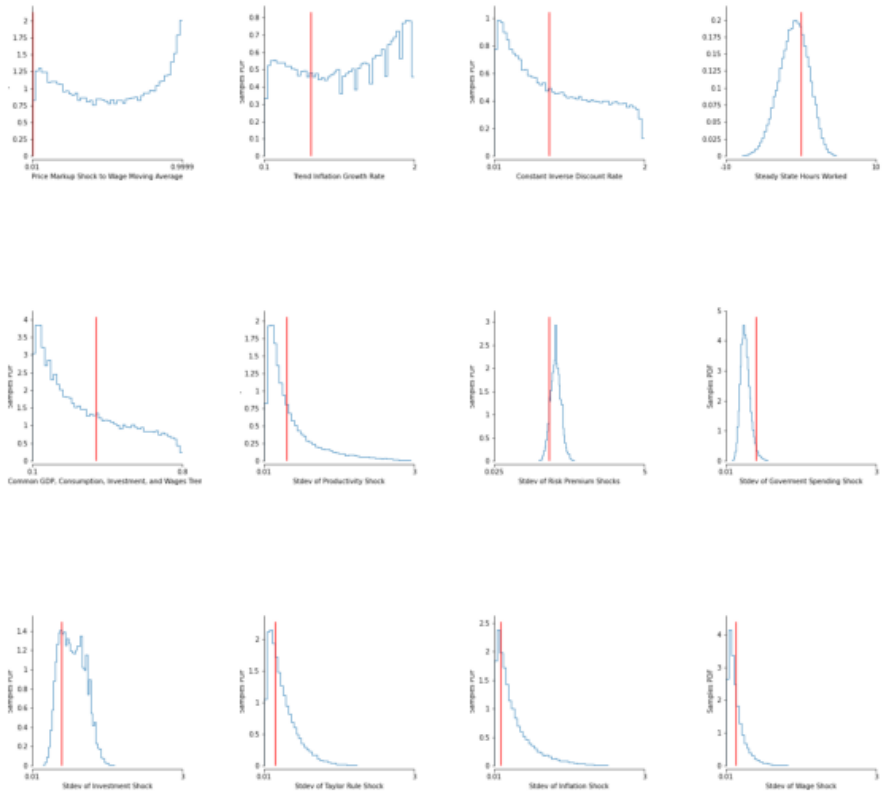

Figure XI: Smets-Wouters Posterior with Parameters 25-36

The Smets-Wouters model demonstrates varying levels of success in estimating the various parameters. For example, the model accurately estimates the capital share but has more difficulty with the adjustments costs elasticity. The (one plus) fixed costs are estimated well. The Taylor rule parameters for output gap and change in output gap are estimated well too, while the inflation Taylor rule parameter is diffuse. Some of the autoregressive parameters for the shocks are well estimated, while others are diffuse or located in areas of low probability mass. The model performs well in estimating the standard deviation of the shock variables.

The posteriors estimated by the Smets-Wouters model with MH-MCMC are less successful in covering the true data generating process parameters. Despite the use of state-of-the-art Dynare software (Adjemian et al., 2022) to draw samples, the model took roughly 30 hours to draw 10

million samples, and the posterior showed limited coverage even after ten thousand sample burn in per chain. Below I reproduce the (Smets and Wouters, 2007) using MH-MCMC instead.

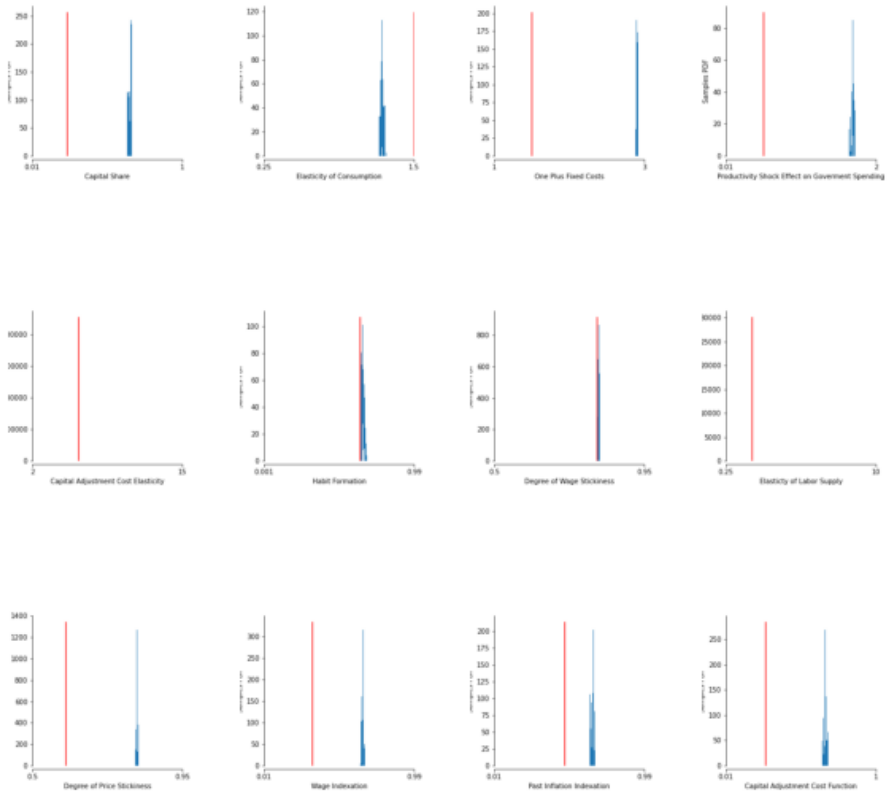

Figure XII: Smets-Wouters MH-MCMC Posterior with Parameters 1-12

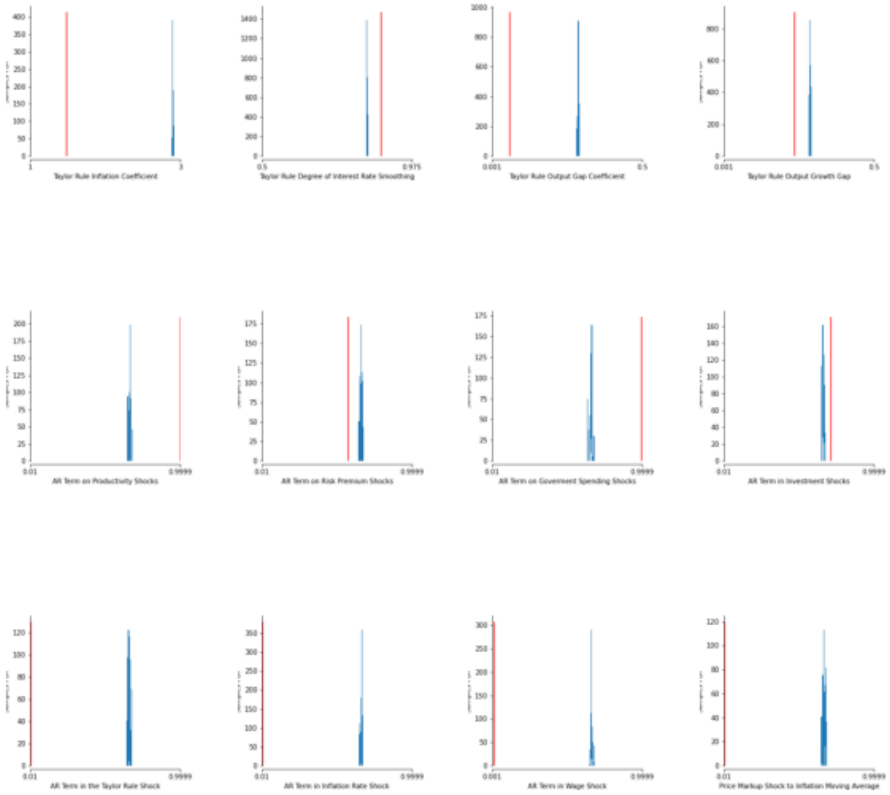

Figure XIII: Smets-Wouters MH-MCMC Posterior with Parameters 13-24

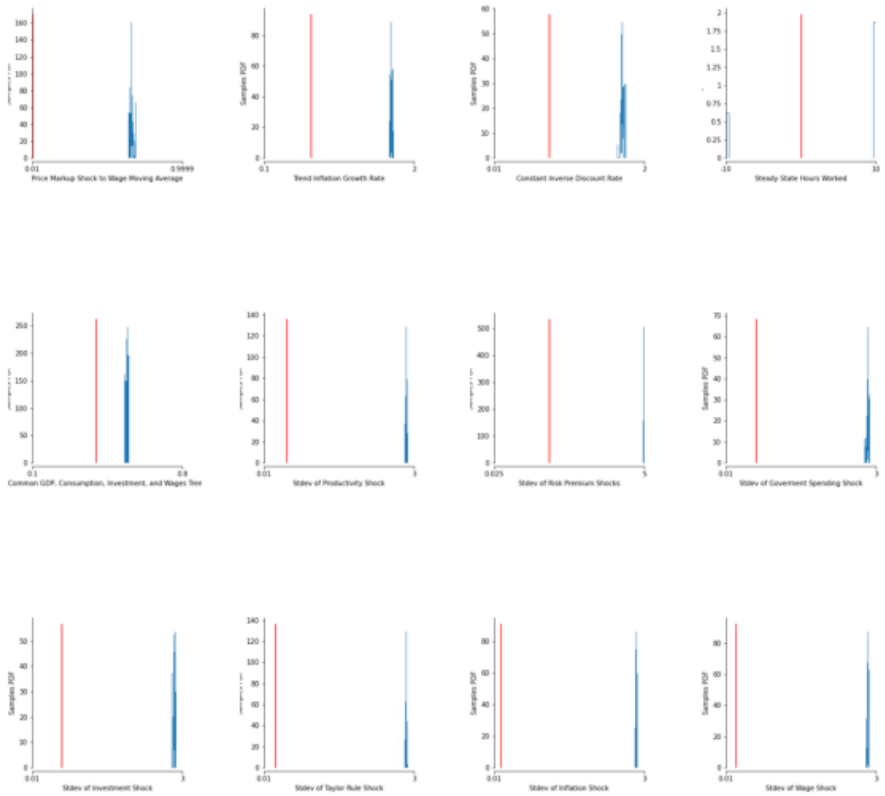

Figure XIV: Smets-Wouters MH-MCMC Posterior with Parameters 25-36

The performance of the MH-MCMC algorithm, as applied in this study, presents several challenges in terms of its ability to converge to the true data generating parameters. While some of the posteriors show good coverage of the true parameters, a substantial number of the posteriors exhibit a support outside of the ground truth and often converge to delta functions far from the true parameters. In particular, the AR term in wage shock equation and price markup shock variables stand out as presenting the most glaring errors, with the MH-MCMC algorithm placing all its mass near .8, whereas the true parameter is located at .015. On the other hand, the SNPE posterior presents a more reasonable representation, putting some mass at .99 and also exhibiting a mode at .015 for both variables. This mistake is repeated to a lesser extent for other variables such as Taylor coefficient on output gap growth, Taylor coefficient for inflation, constent trend inflation

growth rate, steady state hours worked, standard deviation of productivity, among many others. The posteriors in SNPE have no issue reproducing multi-modal posteriors, a documented problem with MH-MCMC and is illustrated in the images Herbst and Schorfheide (2014).

The poor convergence of the MH-MCMC algorithm is believed, in part, to be due to the use of uninformative priors, especially those that are independent across marginal dimensions. Such priors structure the posterior in a way that makes it more independent across marginal dimensions, which, in turn, allows the marginally independent random walk kernel used in the MH-MCMC to better explore the posterior. In contrast, uninformative priors do not influence the posterior to be more independent across marginal dimensions, resulting in strong covariance structures across axes, making the task of finding good proposal samples more challenging for the marginally independent random walk kernel. Even in the cases of informational priors, there is much evidence for the lack of convergence under realistic computational resources for MH-MCMC. For example, Herbst and Schorfheide (2014) demonstrate the identification problem for MH-MCMC in the Smets-Wouters model, while Andrews and Mikusheva (2015) highlight theoretical problems in identifying DSGE models in the MLE setting. When using MH-MCMC simulated annealing to perform MLE optimization, Siarry et al. (1997) demonstrate curse of dimensionalty issues even with objective functions with as few as 3-6 parameters

The SNPE algorithm used in this study was implemented in Python, while the MH-MCMC algorithm was performed in Dynare with pre-loaded Smets-Wouters code from the authors, with only the priors changed to uniform priors and some estimation settings modified. However, due to the difference in programming languages and the overhead of calling Matlab code in Python, the estimation process was much slower for the SNPE algorithm, with roughly 2 models solved per second, compared to about 100 models solved per second for the MH-MCMC. If the solution method in both algorithms were written in Python, MH-MCMC would be significantly slower than it is now. Tests indicate that MH-MCMC has a 3x to 5x advantage in terms of model solution speed Nevertheless, to ensure a fair comparison, the SNPE posterior is shown below after 150 thousand samples, which took roughly the same 30 hours as the 10 million samples of the MH-MCMC:

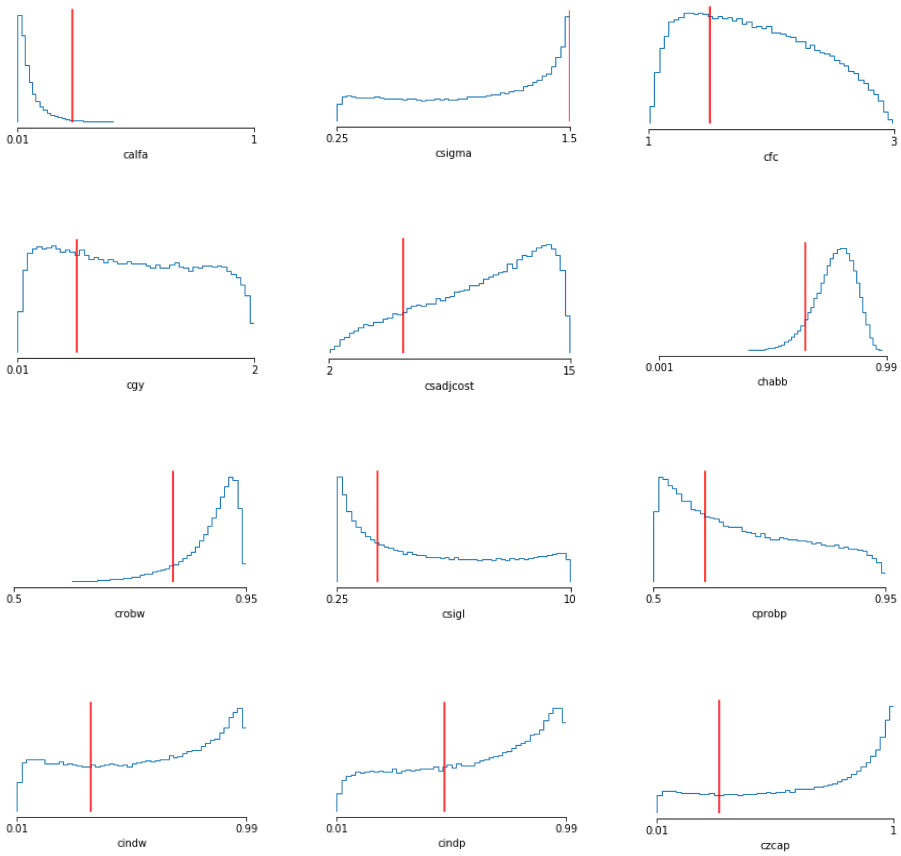

Figure XV: Smets-Wouters SBI Posterior with 150 Thousand Samples: Parameters 1-12

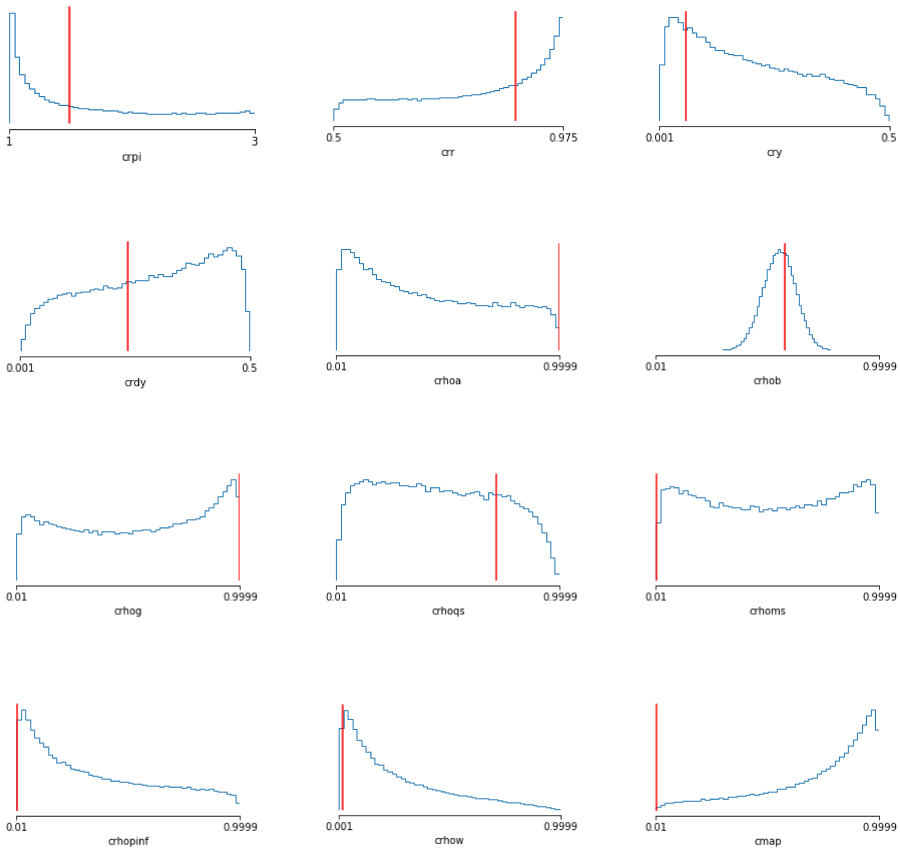

Figure XVI: Smets-Wouters SBI Posterior with 150 Thousand Samples: Parameters 13-24

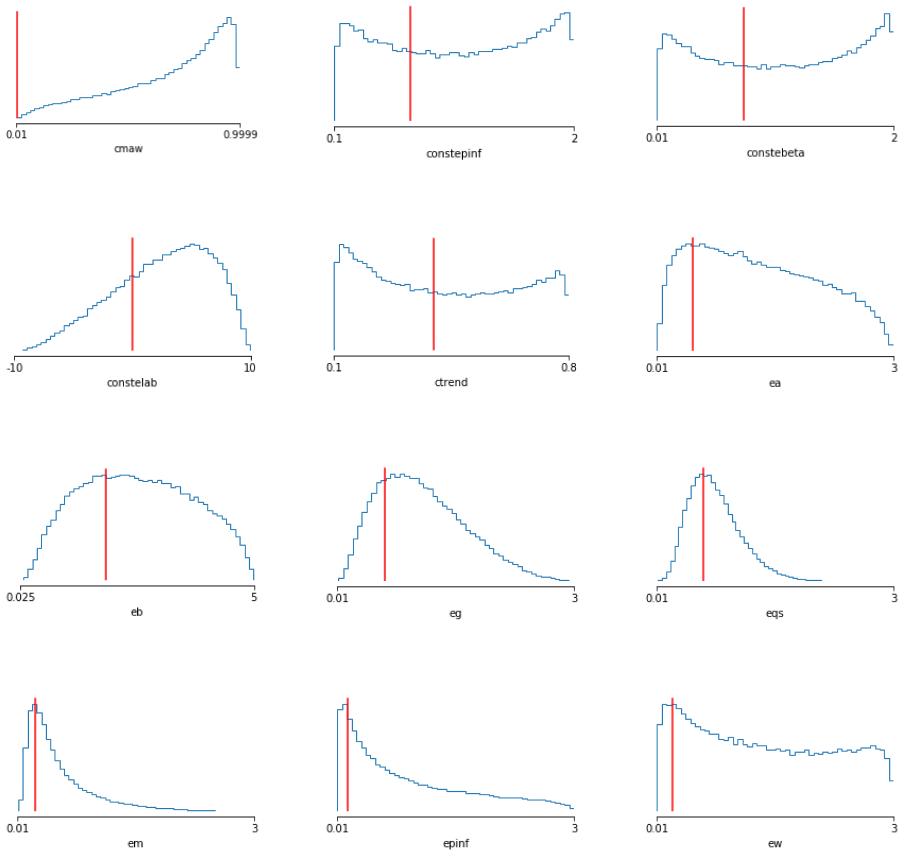

Figure XVII: Smets-Wouters SBI Posterior with 150 Thousand Samples: 25-36

Despite the slower sampling speed of the Python-based SNPE algorithm, it still yields a more accurate posterior compared to the MH-MCMC algorithm. Although the SNPE algorithm incurs a time penalty, it still manages to learn a better posterior in the same amount of time as the MH-MCMC. The performance of the SNPE algorithm can be further improved by optimizing the Python code through techniques such as numba or writing the backend algorithm in C++, however, these are beyond the scope of this paper. The results indicate that SNPE posterior is not perfect, however, it still assigns a significant probability mass to the true data-generating process and in some cases, even concentrates around the true parameters. This is a marked improvement compared to the MH-MCMC algorithm, which even after 10 million iterations, generally only produces highly

concentrated posteriors along a small range that often does not include the true data-generating parameter.

IV.E. HANK Model

The fifth model is the 10 parameter HANK model with the parameters discussed in the model section:

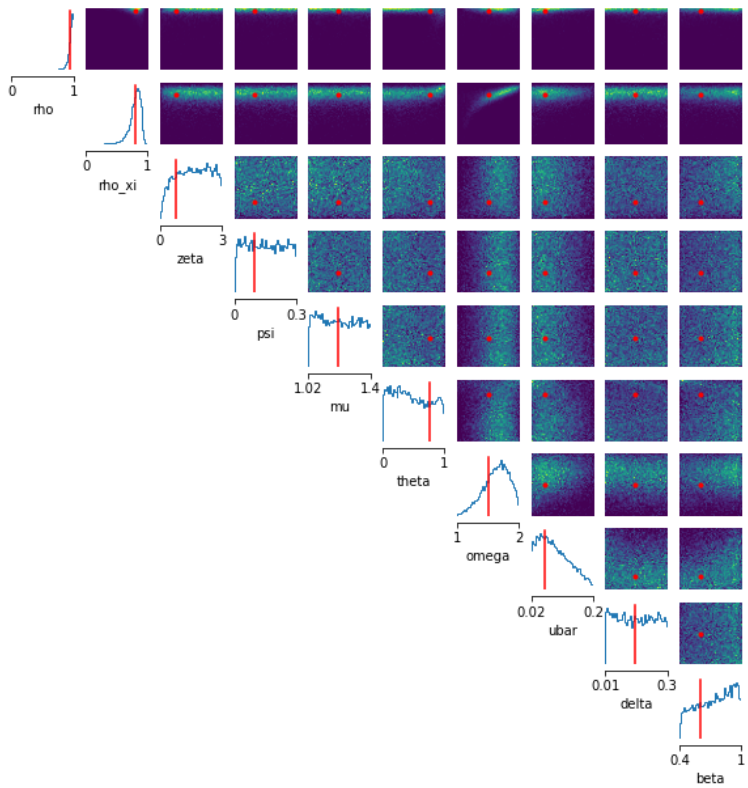

Figure XVIII: Ten Parameter HANK model–Partially Observed

The estimation of the HANK model is the focus of the next discussion. The HANK model features two observable shocks, namely the unemployment rate and the interest rate, and thus only two observable equations and was solved with Reiter's method (Reiter, 2009). Despite having a large latent state space of 810 latent variables, the estimation of the model was carried out successfully removing the need for lossy dimesnionality reduction of Ahn et al. (2018). Dimensionality reduction

is needed when evaluating the likelihood as filtering rapidly runs into the curse of dimensionality. However, my approach doesn't require likelihood evaluations and by extension the intermediate filtering step. As one can see, the approach used in the estimation process was able to identify the parameters with confidence, and the estimated parameters were found to be in agreement with the ground truth. This demonstrates the feasibility of estimating HANK models with a large latent state space, which is even computationally infeasible for an approach with a likelihood without using dimension reduction.

The approach used in the estimation process was able to estimate the parameters ρ, ρ_ξ, ω and $\bar{u}$ with high accuracy. The mode of the distribution of these parameters was found to be either at the true parameter value or in the case of ω, close to the true parameter value. However, for the other parameters, the likelihoods were found to be flat, which may be attributed to the limited impact of these parameters on the chosen observed variables. In conclusion, the estimation of the HANK model showed promising results, and highlights the potential of the approach used in the estimation process to estimate models with large latent state spaces.

IV.F. Winberry Heterogenous Agent Model

A sixth model is a Winberry (2018) approach to the Krusell and Smith (1998) model with full micromoments information as implemented in Liu and Plagborg-Møller (2023). My approach recovers similiar posterior results as Liu and Plagborg-Møller (2023), but my approach is around 13x faster per iteration as I only need micro and macro variable samples, without having the reconstruct the likelihood and use a sampling approach to numerically approximate an integral. I use the same number of iterations (10000) as Liu and Plagborg-Møller (2023), and seems to converge to the posterior like Liu and Plagborg-Møller (2023). I attempted to estimate even more parameters and even use empirical data, however, the results suggest the model is unidentified, so I did not show my results. Liu and Plagborg-Møller (2023) do not show estimation results with more parameters and with empirical rather than simulated data as well.

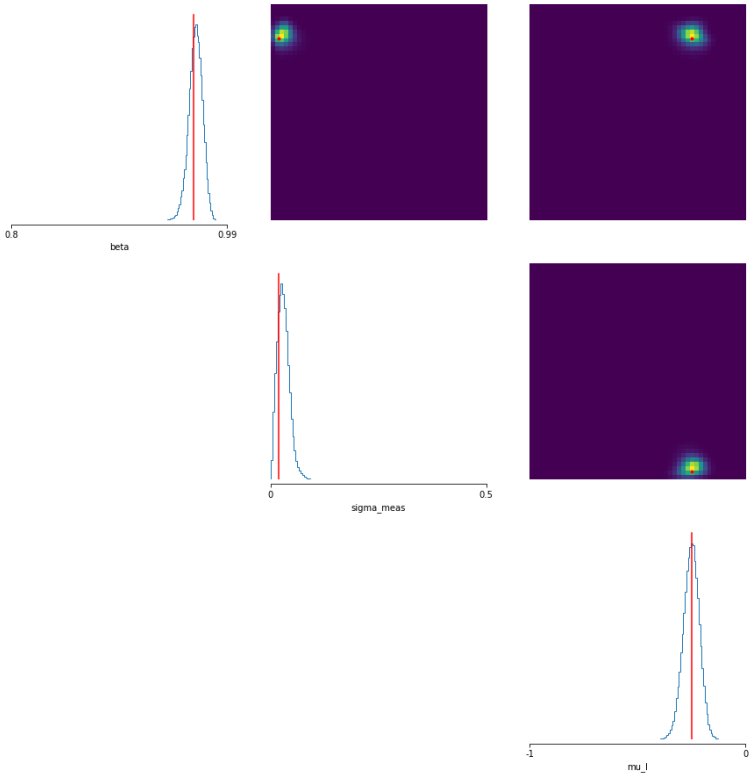

Figure XIX: SBI Results

Here are the Liu and Plagborg-Møller (2023) results:

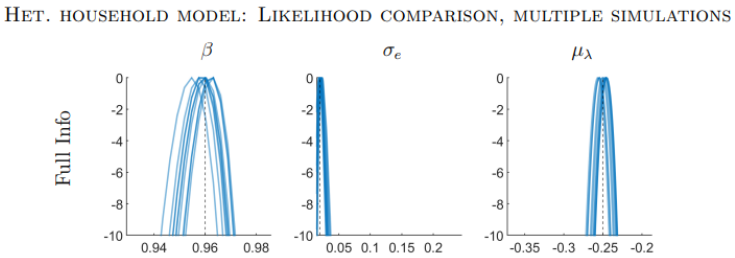

Figure XX: Liu and Plagborg-Moller Results

The results of the study are presented in the form of a figure above. The figure shows that the posterior concentrates almost entirely around the true parameter value, suggesting that the estimation was highly informative regarding the parameter value and accurately reproduces the

simulation parameters. The results are compared to those obtained by Liu and Plagborg-Møller (2023), which are presented in above as well. Both sets of results are similar and demonstrate the viability of the approach used in this study. This example illustrates how SNPE is a fast and reliable approach to the estimation of heterogeneous agent macroeconomic models.

Additionally, using the Winberry solution method, and extending Liu and Plagborg-Møller (2023), I estimated Krusell and Smith (1998) model on empirical data using both micro wealth quantiles, and macro variables: output, investment, consumption and wage. The estimation routine results in posteriors that are somewhat diffuse, and if Liu and Plagborg-Møller (2023) attempted estimation with MH-MCMC, it is likely the approach returned different posterior samples depending on random seed and starting location. However, SNPE produces reproducible posteriors, even when they are flat, because it doesn't depend on a random walker exploring the posterior. The variables are β: the discount rate; μ_l: the wealth heterogeneity parameter; measurement error for each of the 4 macro varaibles and the micro wealth process (meas_error5); and a lower bound on net borrowing.

The macro data comes from standard source (FRED etc.). The micro data comes from the total household net worth variable in the Survey of Income and Program Participation (SIPP). Although not all years are recorded, the years where the participant wealth was provided was transformed into the first to 99th quantiles. At the same time, the Krussel Smith model also produced both simulated macro data and micro wealth quantiles. Here is the posterior data:

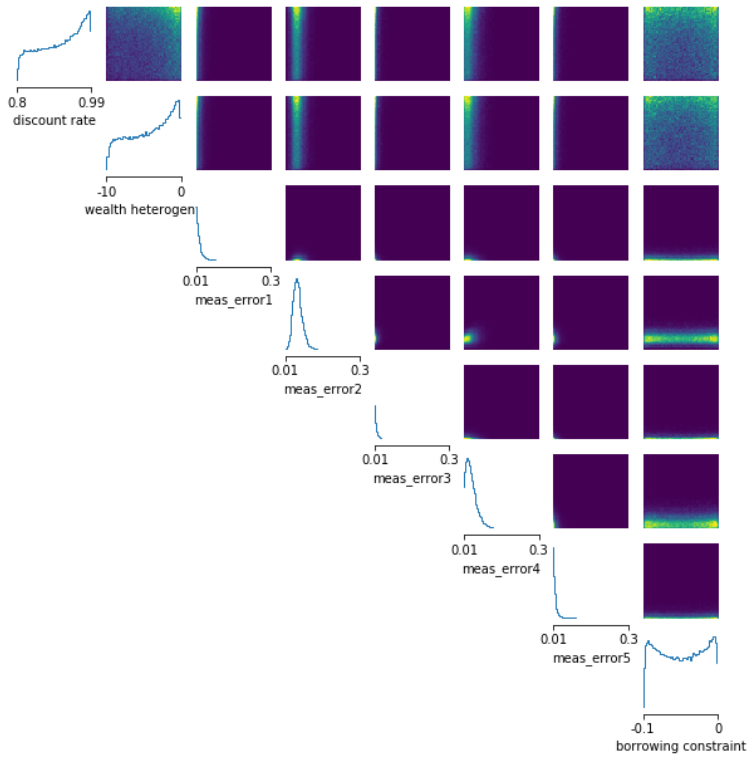

Figure XXI: Krussel Smith/Winberry Empirical Results

IV.G. Bequests Model

The seventh model is the 11 parameter bequests model with parameters displayed in the chart corresponding to the parameters in the model appendix section:

The seventh model considered in this study is an 11-parameter bequests model. This chart corresponds to the parameters in the model appendix section of the paper.

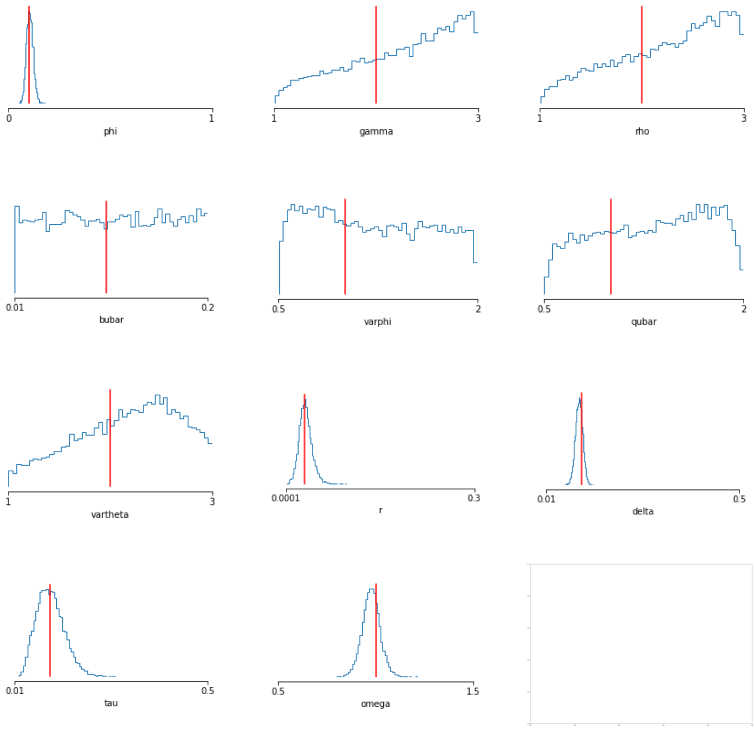

Figure XXII: Eleven Parameter Model Estimated via VFI

The results of the estimation of the model parameters via value function iteration (VFI) are shown in the figure. Although many of the parameters have converged to the true values, some of the parameters still exhibit high uncertainty. This uncertainty could be due to the fact that the parameters have very flat likelihoods. Many of the posterior distributions do concentrate around the true values, suggesting that the model has some discerning power with regard to the parameters.

V. Empirical Application

Not done yet...

VI. Conclusion

In this paper, I introduce a novel approach to estimate models in a Bayesian or MLE framework, which does not require a likelihood function and also derives posteriors faster and more accurately than baseline MH-MCMC and other competing methods. This approach enables the estimation of models while preserving the state space structure of the data generating process. The method has been applied to HANK models with large latent state spaces and has been successful in providing credible posterior estimates. Moreover, it can be applied to solution methods such as value function iteration and projection, which do not yield a likelihood function. This approach can be applied as broadly as MSM with improved guarantees for efficiency and posterior estimation.

The proposed method has a wider range of applications beyond macroeconomics. It can be used in the fields of industrial organization, applied game theory, and econometrics. In industrial organization and game theory, the method can be used to solve structural and dynamic models. In econometrics, it can be applied to estimate general latent time series models effectively and efficiently, without the need for a likelihood function. Additionally, the the suite of conditional density estimators can be used for density estimation in high dimensional settings.

In conclusion, the proposed method has a much broader scope of application than just macroeconomic modeling. The applications of this technique are not limited to this paper, and have the potential to be explored in greater depth in future research.

References

Adjemian, Stéphane, Houtan Bastani, Michel Juillard, Fréderic Karamé, Ferhat Mihoubi, Willi Mutschler, Johannes Pfeifer, Marco Ratto, Normann Rion, and Sébastien Villemot. 2022. "Dynare: Reference Manual Version 5." CEPREMAP Dynare Working Papers 72.

Ahn, SeHyoun, Greg Kaplan, Benjamin Moll, Thomas Winberry, and Christian Wolf. 2018. "When inequality matters for macro and macro matters for inequality." *NBER macroeconomics annual*, 32(1): 1–75.

Andrews, Isaiah, and Anna Mikusheva. 2015. "Maximum likelihood inference in weakly identified dynamic stochastic general equilibrium models." *Quantitative Economics*, 6(1): 123–152.

Auclert, Adrien, Bence Bardóczy, Matthew Rognlie, and Ludwig Straub. 2021*a*. "Using the sequence-space Jacobian to solve and estimate heterogeneous-agent models." *Econometrica*, 89(5): 2375–2408.

Auclert, Adrien, Hannes Malmberg, Frederic Martenet, and Matthew Rognlie. 2021*b*. "Demographics, wealth, and global imbalances in the twenty-first century." National Bureau of Economic Research.

Bishop, Christopher M. 1994. "Mixture density networks."

Boelts, Jan, Jan-Matthis Lueckmann, Richard Gao, and Jakob H Macke. 2021. "Flexible and efficient simulation-based inference for models of decision-making." *bioRxiv*.

Brehmer, Johann. 2021. "Simulation-based inference in particle physics." *Nature Reviews Physics*, 3(5): 305–305.

Caballero, Ricardo J, and Eduardo MRA Engel. 2007. "Price stickiness in Ss models: New interpretations of old results." *Journal of monetary economics*, 54: 100–121.

Caglio, Cecilia R, R Matthew Darst, and Sebnem Kalemli-Özcan. 2021. "Risk-taking and monetary policy transmission: Evidence from loans to smes and large firms." National Bureau of Economic Research.

Chewi, Sinho. 2023. "Log Concave Sampling."

Cranmer, Kyle, Johann Brehmer, and Gilles Louppe. 2020. "The frontier of simulation-based inference." *Proceedings of the National Academy of Sciences*, 117(48): 30055–30062.

DiNapoli, Robert J, Enrico R Crema, Carl P Lipo, Timothy M Rieth, and Terry L Hunt. 2021. "Approximate Bayesian Computation of radiocarbon and paleoenvironmental record shows population resilience on Rapa Nui (Easter Island)." *Nature Communications*, 12(1): 1–10.

Dixit, Avinash K, and Joseph E Stiglitz. 1977. "Monopolistic competition and optimum product diversity." *The American economic review*, 67(3): 297–308.

Duffie, Darrell, and Kenneth J Singleton. 1990. "Simulated moments estimation of Markov models of asset prices."

Durkan, Conor, Iain Murray, and George Papamakarios. 2020. "On contrastive learning for likelihood-free inference." 2771–2781, PMLR.

Farrell, Max H, Tengyuan Liang, and Sanjog Misra. 2018. "Deep neural networks for estimation and inference: Application to causal effects and other semiparametric estimands." *arXiv preprint arXiv:1809.09953*.

Fernández-Villaverde, Jesús, Juan Francisco Rubio-Ramírez, and Frank Schorfheide. 2016. "Solution and estimation methods for DSGE models." In *Handbook of macroeconomics*. Vol. 2, 527–724. Elsevier.

Gallant, A Ronald, and George Tauchen. 1996. "Which moments to match?" *Econometric theory*, 12(4): 657–681.

Glöckler, Manuel, Michael Deistler, and Jakob H Macke. n.d.. "Variational methods for simulation-based inference."

Goodfellow, Ian, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. 2014. "Generative adversarial nets." *Advances in neural information processing systems*, 27.

Greenberg, David, Marcel Nonnenmacher, and Jakob Macke. 2019. "Automatic posterior transformation for likelihood-free inference." 2404–2414, PMLR.

Haan, Peter, and Arne Uhlendorff. 2006. "Estimation of multinomial logit models with unobserved heterogeneity using maximum simulated likelihood." *The Stata Journal*, 6(2): 229–245.

Hajivassiliou, Vassilis A, et al. 1997. *Some practical issues in maximum simulated likelihood.* Suntory-Toyota International Centre for Economics and Related Disciplines ....

Hansen, Lars Peter. 1982. "Large sample properties of generalized method of moments estimators." *Econometrica: Journal of the econometric society*, 1029–1054.

Hennessy, Christopher A, and Toni M Whited. 2007. "How costly is external financing? Evidence from a structural estimation." *The Journal of Finance*, 62(4): 1705–1745.

Herbst, Edward, and Frank Schorfheide. 2014. "Sequential Monte Carlo sampling for DSGE models." *Journal of Applied Econometrics*, 29(7): 1073–1098.

Herbst, Edward P, and Frank Schorfheide. 2015. *Bayesian estimation of DSGE models.* Princeton University Press.

Hermans, Joeri, Volodimir Begy, and Gilles Louppe. 2020. "Likelihood-free mcmc with amortized approximate ratio estimators." 4239–4248, PMLR.

Huang, Chin-Wei, David Krueger, Alexandre Lacoste, and Aaron Courville. 2018. "Neural autoregressive flows." 2078–2087, PMLR.

Judd, Kenneth L, Lilia Maliar, Serguei Maliar, and Inna Tsener. 2017. "How to solve dynamic stochastic models computing expectations just once." *Quantitative Economics*, 8(3): 851–893.

Kaji, Tetsuya, Elena Manresa, and Guillaume Pouliot. 2020. "An adversarial approach to structural estimation." *arXiv preprint arXiv:2007.06169.*

Keane, Michael P, and Kenneth I Wolpin. 1994. "The solution and estimation of discrete choice dynamic programming models by simulation and interpolation: Monte Carlo evidence." *the Review of economics and statistics*, 648–672.

Khan, Aubhik, and Julia K Thomas. 2008. "Idiosyncratic shocks and the role of nonconvexities in plant and aggregate investment dynamics." *Econometrica*, 76(2): 395–436.

Kristensen, Dennis, and Yongseok Shin. 2012. "Estimation of dynamic models with nonparametric simulated maximum likelihood." *Journal of Econometrics*, 167(1): 76–94.

Krusell, Per, and Anthony A Smith, Jr. 1998. "Income and wealth heterogeneity in the macroeconomy." *Journal of political Economy*, 106(5): 867–896.

Kukackaa, Jiri, and Jozef Barunika. n.d.. "Non-Parametric Simulated Maximum Likelihood Estimation of Parameters in Heterogeneous Agent ModelsI Preliminary Draft Version: May 19, 2015."

Lerman, Steven, and Charles Manski. 1981. "On the use of simulated frequencies to approximate choice probabilities." *Structural analysis of discrete data with econometric applications*, 10: 305–319.

Liu, Laura, and Mikkel Plagborg-Møller. 2023. "Full-information estimation of heterogeneous agent models using macro and micro data." *Quantitative Economics*, 14(1): 1–35.

Lucas Jr, Robert E. 1978. "Asset prices in an exchange economy." *Econometrica: journal of the Econometric Society*, 1429–1445.

McFadden, Daniel. 1989. "A method of simulated moments for estimation of discrete response models without numerical integration." *Econometrica: Journal of the Econometric Society*, 995–1026.

Mohammad-Djafari, Ali, and Hacheme Ayasso. 2009. "Variational Bayes and mean field approximations for Markov field unsupervised estimation." 1–6, IEEE.

Ottonello, Pablo, and Thomas Winberry. 2020. "Financial heterogeneity and the investment channel of monetary policy." *Econometrica*, 88(6): 2473–2502.

Pakes, Ariel, and David Pollard. 1989. "Simulation and the asymptotics of optimization estimators." *Econometrica: Journal of the Econometric Society*, 1027–1057.

Papamakarios, George, and Iain Murray. 2016. "Fast ε-free inference of simulation models with bayesian conditional density estimation." *Advances in neural information processing systems*, 29.

Parra-Alvarez, Juan Carlos, Olaf Posch, and Mu-Chun Wang. 2020. "Estimation of heterogeneous agent models: A likelihood approach."

Reiter, Michael. 2009. "Solving heterogeneous-agent models by projection and perturbation." *Journal of Economic Dynamics and Control*, 33(3): 649–665.

Rendahl, Pontus. 2017. "Linear time iteration." IHS Economics Series.

Rezende, Danilo, and Shakir Mohamed. 2015. "Variational inference with normalizing flows." 1530–1538, PMLR.

Siarry, Patrick, Gérard Berthiau, François Durdin, and Jacques Haussy. 1997. "Enhanced simulated annealing for globally minimizing functions of many-continuous variables." *ACM Transactions on Mathematical Software (TOMS)*, 23(2): 209–228.

Smets, Frank, and Rafael Wouters. 2007. "Shocks and frictions in US business cycles: A Bayesian DSGE approach." *American economic review*, 97(3): 586–606.

Terry, Stephen J. 2017. "Alternative methods for solving heterogeneous firm models." *Journal of Money, Credit and Banking*, 49(6): 1081–1111.

Tovar, Camilo E. 2009. "DSGE models and central banks." *Economics*, 3(1).

Winberry, Thomas. 2018. "A method for solving and estimating heterogeneous agent macro models." *Quantitative Economics*, 9(3): 1123–1151.

A Adaptability of SNPE to Estimation of Models Solved via a Variety of Methods

Solution Method/Algorithm	**Details**	**SNPE**	**MCMC**	**SMC**	**MSM**
Vanilla Perturbation	Posterior Quality	✓	∼	✓	✗
	Ability to Estimate	✓	✓	✓	∼
Vanilla Projection	Posterior Quality	✓	∼	✓	✗
	Ability to Estimate	✓	∼	∼	∼
Value Function Iteration	Posterior Quality	✓	∼	✓	✗
	Ability to Estimate	∼	∼	∼	∼
HANK Reiter	Posterior Quality	✓	∼	∼	✗
	Ability to Estimate	✓	∼/✗	∼/✗	∼
HANK Winberry	Posterior Quality	✓	∼	∼/?	✗
	Ability to Estimate	✓	∼	∼/?	∼

B Convergence of the Embedding Network to the Sufficient Statistics Function

Assumptions: Each set of R^n parameters generates a single distribution, X from the model. Asymptotically the set of X grows and is iid with respect to a hidden state. One set of parameters cannot generate two different distributions. Likewise the maximum likelihood regularity assumptions for posterior density estimation also hold, although since we aren't after parameter identification that does not need to hold. However boundedness of the likelihood function, differentionability of the log likellihood and uniform intergralbility all need to hold, for the density estimator that takes samples θ, X and estimates the conditional $P(\theta|X^*)$ which ultimately produces the posterior when conditioned on X, $P(\theta|X^* = X)$.

Proof: As X goes to infinity, and as the number of samples $P(\theta, X)$ goes to infinity, one can

do no better than an end-to-end optimizable unviversal approximator embedding that reduces the dimensionality of X to a set vector of size n. For any distribution that can be generated by theta, there is one unique theta that generates it. Thus there exists a sufficient statistic of size theta, no matter the size of X. When estimating the conditional distribution $P(\theta|X)$ via maximum likelihood, as the size of X increases, as X is iid conditional on the hidden state, the statistic functions $\hat{g}(X)$ converge to their population counterparts $g(X)$, via the law of large numbers. The parametric MLE density estimator $\hat{P}(\theta|X)$ will converge to the true distribution $P(\theta|X)$ asymptotically in X, θ samples, via the consistency of MLE as long as the density estimator is a universal approximator. Likewise by Rao-Blackwell, one can do no better than estimating the density via the embedding $g^*(x)$ in $\hat{P}(\theta|g^*(X))$ assuming $g^*(x)$ is a sufficient statistic function. Since one wants to maximize the likelihood of $\hat{P}(\theta|X)$, the optimal function $g(X)$ is $g^*(X)$, the sufficient statistic function. Thus the embedding feed forward neural network, given enough θ, X samples, as a universal approximator, will converge to a function that performs no worse than $g^*(X)$, the sufficient statistic function. And $\hat{P}(\theta|g(X))$ will converge to something equivalent to the population conditional distribution: $P(\theta|g^*(X))$.

C Economic Models

The models used to test these algorithms are disparate. I estimate a relatively standard RBC model via time iteration, a model of capital formation solved via value function iteration, a Lucas Tree model (Lucas Jr, 1978) solved with projection, the linearized Smets-Wouters 2007 model via perturbation, a HANK model solved via Rieter's method with time iteration and lastly a 11 parameter bequests model solved via value function iteration. I will now discuss each of these models sequentially.

CA. The RBC Model

First I will estimate an RBC model. This RBC model is standard and comes from Alisdair McKay's note on heterogeneous agent models: `https://alisdairmckay.com/Notes/HetAgents/`

`index.html`.

The utility function is a standard CRRA utility:

$$u(C_t) = E \sum_t \beta^t \frac{C_t^{1-\gamma}}{1-\gamma} \tag{1}$$

The production function is standard Cobb-Douglas with productivity:

$$Y_t = Z_t K_t^{\alpha} \bar{L}^{1-\alpha} \tag{2}$$

Where $\bar{L}$ is fixed labor and Z_t is productivity and K_t is capital which evolves according to a standard law of motion:

$$K_t = (1-\delta)K_{t-1} + Y_t - C_t \tag{3}$$

Productivity evolves according to a AR(1) process in logs:

$$logZ_t = \rho logZ_{t-1} + \epsilon_t \tag{4}$$

This model is solved via time iteration (Rendahl, 2017) and matches consumption, investment, productivity to real data, interest rates and capital are the latent unobserved variables. This is done on data both synthetic and real.

CB. The Model of Capital Formation

Next I will describe a partial equilibrium model of capital formation. This model is a straightforward infinite horizon partial equilibrium model with risk neutral managers choosing investment to maximize the present value of a stream of firm cash flows and comes from solution code written by Emil Lakkis and Toni Whited.

Utility/cash flows are give by this formula:

$$U(K_t, I_t, Z_t) = \sum_t \beta^t * (\pi(K_t, Z_t) - I_t) \tag{5}$$

Where K_t, Z_t, and I_t are the same as in the RBC model–capital, productivity and investment respectively. π is the profit function.

The capital law of motion is like the RBC model:

$$K_t = (1 - \delta)K_{t-1} + I_t \tag{6}$$

Productivity evolves according to a AR(1) process in logs:

$$logZ_t = \rho logZ_{t-1} + \epsilon_t \tag{7}$$

Profit is defined as:

$$\pi(K_t, Z_t) = Z_t K_t^{\alpha} \tag{8}$$

This model shares similarities with the RBC model, with the main difference being the utility function and the value function solution method. This estimation method matches capital levels generated by a parameterized model.

CC. Lucas Asset Pricing Projection Model

' Next I will estimate a Lucas Tree model solved via projection. Following Mohammed Ait Lahren's code (`https://notes.quantecon.org/submission/5f44808ad24fdb001162a53b`), each agent maximizes CRRA utility:

$$max_i E_t[\beta^t \frac{c_t^{1-\gamma} - 1}{1 - \gamma} \tag{9}$$

with γ set to 3. The law of motion is:

$$k_{t+1} = (1 + d_t/P_t)k_t - c_t/P_t \tag{10}$$

The state of the system is defined as dividends received plus price of the amount of assets owned:

$$m_{t+1} = (P_{t+1} + d_{t+1})k_{t+1} \tag{11}$$

Putting these three equations together one gets the pricing kernel:

$$p_t = \beta^t E_t[(\frac{c_{t+1}}{c_t})^{-\gamma}(d_{t+1} + p_{t+1})] \tag{12}$$

Dividends evolve under an $AR(1)$ process:

$$d_{t+1} = \mu_d + \rho_d * d_t + \epsilon \tag{13}$$

Where ϵ is a shock with standard deviation σ. When behaving optimally, consumption also equals the dividends received. This model matches the expected return and risk free rate at a variety of dividend levels, which demonstrates the ability of this model to not only estimate on models solved via projection, but also to work on cross sectional problems as well.

CD. Smets-Wouters 2007 Model

Additionally I will estimate a Smets Wouters 2007 model (Smets and Wouters, 2007). This section will mainly illustrate the equations and the parameters so one can cross reference parameters shown in the diagram with the correct equation in the DSGE model. This model comes from the paper Smets and Wouters (2007). More details on the economics of the equations can be found there. This model can be estimated both with MH-MCMC and SNPE. Thus this model mostly

illustrates the increases in posterior accuracy when using an uninformative prior which goes hand in hand with estimation speed when using SNPE vs MH-MCMC. Here is a reference for all the Smets-Wouter abbreviations and the actual variable name:

calfa = Capital Share

csigma = Elasticity of Consumption

cfc = One Plus Fixed Costs

cgy = 'Productivity Shock Effect on Government Spending

csadjcost = Capital Adjustment Cost Elasticity

chabb = Habit Formation

cprobw = Degree of Wage Stickiness

csigl = Elasticity of Labor Supply

cprobp = Degree of Price Stickiness

cindw = Wage Indexation cindp = 'Past Inflation Indexation

czcap = Capital Adjustment Cost Function

crpi = Taylor Rule Inflation Coefficient

crr = Taylor Rule Degree of Interest Rate Smoothing

cry = Taylor Rule Output Gap Coefficient

crdy = 'Taylor Rule Output Growth Gap Coefficient

crhoa = AR Term on Productivity Shocks

crhob = AR Term on Risk Premium Shocks

crhog = 'AR Term on Government Spending Shocks

crhoqs = .AR Term in Investment Shocks

crhoms = AR Term in the Taylor Rule Shock

crhopinf = AR Term in Inflation Rate Shock

crhow = AR Term in Wage Shock

cmap = 'Price Markup Shock to Inflation Moving Average

cmaw = Price Markup Shock to Wage Moving Average

constepinf ='Trend Inflation Growth Rate

constebeta = Constant Inverse Discount Rate

constelab = Steady State Hours Worked

ctrend = Common GDP, Consumption, Investment, and Wages Trend

ea =Stdev of Productivity Shock

eb =Stdev of Risk Premium Shocks

eg = 'Stdev of Government Spending Shock

eqs =Stdev of Investment Shock

em ='Stdev of Taylor Rule Shock

epinf = Stdev of Inflation Shock

ew = Stdev of Wage Shock

First, a brief discussion of the overview of the model. The model is a representative New Keynesian model with stickiness in both wages and prices. The model has a flexible price set of equations which determines potential GDP for monetary policy and a sticky price system with a price markup equation that determines the true economy. I will only be discussing the sticky price system of equations with an implicit understanding that there is a mirroring flexible price system. Shocks all have autoregressive components to more accurately match data. All equations are linearized.

The aggregate resource constraint is:

$$y_t = ccy * c_t + ciy * i_t + \epsilon_t^g + 1 * crkky * z_t \tag{14}$$

The Euler equation is given by:

$$\begin{aligned} c_t =& (chabb/cgamma)/(1 + chabb/cgamma) * c_{t-1} + (1/(1 + chabb/cgamma)) * c_{t+1} + \\ & ((csigma - 1) * cwhlc/(csigma * (1 + chabb/cgamma))) * (l_t - l_{t+1}) - \\ & (1 - chabb/cgamma)/(csigma * (1 + chabb/cgamma)) * (r_t - \pi_{t+1}) + \epsilon_t^b \end{aligned} \tag{15}$$

The investment Euler equations are:

(16)

$$i_t = (1/(1+cbetabar*cgamma))*(i_{t-1}+cbetabar*cgamma*i_{t+1}+(1/(cgamma^2*csadjcost))*q_t)+\epsilon_t^i$$

The arbitrage equation for the value of the capital stock is:

$$\begin{aligned} q_t = &- r_t + \pi_{t+1} + (1/((1-chabb/cgamma)/(csigma*(1+chabb/cgamma))))*\epsilon_t^b + \\ &(crk/(crk+(1-ctou)))*r_{t+1}^k + ((1-ctou)/(crk+(1-ctou)))*q_{t+1} \end{aligned} \tag{17}$$

The linearized production function is:

$$y_t = cfc*(calfa*k_t + (1-calfa)*l_t + \epsilon_t^a) \tag{18}$$

Capital utilization:

$$k_t^s = k_{t-1} + z_t \tag{19}$$

Capital utilization is some function of the interest rate for capital:

$$z_t = (1/(czcap/(1-czcap)))*r_t^k \tag{20}$$

Capital's law of motion is:

$$k_t = (1-cikbar)*k_{t-1} + cikbar*i_t + cikbar*cgamma^2*csadjcost*\epsilon_t^i \tag{21}$$

The markup equation is:

$$\mu_t^p = calfa*k_t^s + (1-calfa)*w_t - \epsilon_t^a \tag{22}$$

This equation is the only set of economic equations that doesn't have a flexible price counterpart.

Note that the observed variable equations also will not have a flexible price counterpart.

The New Keynesian Phillips curve:

(23)

$$\pi_t = (1/(1 + cbetabar * cgamma * cindp)) * (cbetabar * cgamma * \pi_{t+1} + cindp * \pi_{t-1} + ((1 - cprobp) * (1 - cbetabar * cgamma * cprobp)/cprobp)/((cfc - 1) * curvp + 1) * (\mu_t^p)) + \epsilon_t^p$$

The rental rate of capital equation:

$$r_t^k = w_t + l_t - k_t \tag{24}$$

The path of wages is:

(25)

$$w_t = (1/(1 + cbetabar * cgamma)) * w_{t-1} + (cbetabar * cgamma/(1 + cbetabar * cgamma)) * w_{t+1} + (cindw/(1 + cbetabar * cgamma)) * \pi_{t-1} - (1 + cbetabar * cgamma * cindw)/(1 + cbetabar * cgamma) * \pi + (cbetabar * cgamma)/(1 + cbetabar * cgamma) * \pi_{t+1} + (1 - cprobw) * (1 - cbetabar * cgamma * cprobw)/((1 + cbetabar * cgamma) * cprobw) * (1/((clandaw - 1) * curvw + 1)) * (csigl * l_t + (1/(1 - chabb/cgamma)) * c_t - ((chabb/cgamma)/(1 - chabb/cgamma)) * c_{t-1} - w_t) + \epsilon_t^w$$

The Taylor rule followed by central banks is:

$$r_t = crpi * (1 - crr) * \pi_t + cry * (1 - crr) * (y_t - y_t^p) + crdy * (y_t - y_t^p - y_{t-1} + y_{t-1}^p) + crr * r_{t-1} + \epsilon_t^r \tag{26}$$

The observed variables equations are:

$$dy_t = y_t - y_{t-1} + ctrend \tag{27}$$

$$dc_t = c_t - c_{t-1} + ctrend \tag{28}$$

$$di_t = i_t - i_{t-1} + ctrend \tag{29}$$

$$dw_t = w_t - w_{t-1} + ctrend \tag{30}$$

$$\pi_t^o = \pi_t + constepinf \tag{31}$$

$$r_t^o = r_t + conster \tag{32}$$

$$l_t^o = l_t + constelab \tag{33}$$

The autoregressive equations for shocks are:

$$\epsilon_t^a = crhoa * \epsilon_{t-1}^a + ea \tag{34}$$

$$\epsilon_t^b = crhob * \epsilon_{t-1}^b + eb \tag{35}$$

$$\epsilon_t^g = crhog * \epsilon_{t-1}^g + eg + cgy * ea \tag{36}$$

$$\epsilon_t^i = crhoqs * \epsilon_{t-1}^i + eqs \tag{37}$$

$$\epsilon_t^r = crhoms * \epsilon_{t-1}^r + em \tag{38}$$

$$\epsilon_t^p = crhopinf * \epsilon_{t-1}^p + epinfma - cmap * epinfma(-1) \tag{39}$$

$$epinfma = epinf \tag{40}$$

$$\epsilon_t^w = crhow * \epsilon_{t-1}^w + ewma - cmaw * ewma(-1) \tag{41}$$

$$ewma = ew \tag{42}$$

Some of the higher level parameters in the above equations are also related to the parameters in my diagram by the following equations:

$$cpie = 1 + constepinf/100 \tag{43}$$

$$cgamma = 1 + ctrend/100 \tag{44}$$

$$cbeta = 1/(1 + constebeta/100) \tag{45}$$

$$clandap = cfc \tag{46}$$

$$cbetabar = cbeta * cgamma^{(-csigma)} \tag{47}$$

$$cr = cpie/(cbeta * cgamma^{(-csigma)}) \tag{48}$$

$$crk = (cbeta^{(-1)}) * (cgamma^{csigma}) - (1 - ctou) \tag{49}$$

$$cw = \frac{(calfa^{calfa} * (1 - calfa)^{(1-calfa)}}{(clandap * crk^{calfa}))^{(1/(1-calfa))}} \tag{50}$$

$$cikbar = (1 - (1 - ctou)/cgamma) \tag{51}$$

$$cik = (1 - (1 - ctou)/cgamma) * cgamma \tag{52}$$

$$clk = ((1 - calfa)/calfa) * (crk/cw) \tag{53}$$

$$cky = cfc * (clk)^{(calfa-1)} \tag{54}$$

$$ciy = cik * cky \tag{55}$$

$$ccy = 1 - cg - cik * cky \tag{56}$$

$$crkky = crk * cky \tag{57}$$

$$cwhlc = (1/clandaw) * (1 - calfa)/calfa * crk * cky/ccy \tag{58}$$

$$cwly = 1 - crk * cky \tag{59}$$

$$conster = (cr - 1) * 100 \tag{60}$$

The seven observed equations dy_t, dc_t, di_t, dw_t, π_t^o, r_t^o, l_t^o, are matched using data simulated from a model with true parameters equal to the Smets-Wouters 2007 mean parameters, with slight modifications as some parameters were calibrated to 0.

CE. The HANK Model

Furthermore, I will estimate a HANK model solved via Reiter's method. This provides an example of estimation of models pushing the frontier in economics. The benefits of this approach allows one to estimate hetergenous agent models in a likelihood manner without resorting to demensionality reduction. The HANK model also comes from Alisdair McKay's tutorial, which adds a heterogeneous distribution of wealth as a state variable for households combined with a relatively standard New Keynesian model.

Preferences are still CRRA, like the RBC model. The firm side is now new Keynesian with intermediate goods produced via:

$$y_{j,t} = Z_t N_{j,t} \tag{61}$$

Here Z_t is aggregate productivity and N_t is labor employed by firm j producing variety $y_{j,t}$. The final good is produced via the Dixit-Stiglitz aggregator Dixit and Stiglitz (1977):

$$Y_t = (\int_0^1 {y_{j,t}}^{\frac{\epsilon - 1}{\epsilon}})^{\frac{\epsilon}{\epsilon - 1}} \tag{62}$$

Here ϵ is the elasticity of substitution between intermediates. Output is defined as:

$$Y_t = A_t \int_0^1 n_{j,t} dj = A_t N_t \tag{63}$$

Where $n_{j,t}$ is the labor for the individual intermediate goods. Additionally from the accounting identity, output is also, $Y_t = C_t + \psi M_t H_t$ where the second term is hiring costs.

Government and bonds are defined by the following nominal and real budget constraint equa-

tions:

$$\tau P_t(w_t + d_t)(1 - u_t) + P_t B = (1 + i_{t-1})P_{t-1}B + P_t b u_t \tag{64}$$

$$\tau(w_t + d_t)(1 - u_t) + B = R_t B + b u_t \tag{65}$$

Labor market is a standard McCall Search model:

$$N_t = (1 - \delta)N_{t-1} + H_t \tag{66}$$

$$1 - u_t = (1 - \delta)(1 - u_{t-1}) + H_t \tag{67}$$

$$M_t = \frac{H_t}{u_{t-1} + \delta N_{t-1}} \tag{68}$$

$$M_t = \frac{1 - u_t - (1 - \delta)(1 - u_{t-1})}{u_{t-1} + \delta(1 - u_{t-1})} \tag{69}$$

The household maximizes the CRRA utility like the RBC model. The households budget constraint in real money is:

$$a_t + C_t = R_t * a_{t-1} + earnings \tag{70}$$

Here a_t is savings and R_t is the interest rate as determined by the bond market clearing condition. Earnings are given by:

$$\text{earnings} = \begin{cases} (1 - \tau_t)\,(w_t + d_t) & \text{if employed} \\ b & \text{if unemployed,} \end{cases} \tag{71}$$

Here w_t is wages, d_t is dividend income and b is unemployment benefits.

The stochastic matrix governing employment, unemployment transitions is given by:

$$\begin{pmatrix} 1 - M_t & \delta \\ M_t & 1 - \delta \end{pmatrix} \tag{72}$$

Again M_t is governed by the labor market equations above.

The firm solves the Dixit-Stiglitz cost problem:

$$y_{j,t} = \left(\frac{p_{j,t}}{P_t}\right)^{-\varepsilon} Y_t \tag{73}$$

with price index P_t given by

$$P_t^{1-\varepsilon} = \int_0^1 p_{j,t}^{1-\varepsilon} dj. \tag{74}$$

The Calvo pricing adjustment parameter θ implies that the intermediate firm chooses price p^* to maximize:

$$\mathbb{E}_t \sum_{s=t}^{\infty} \theta^{s-t} R_{t,s}^{-1} \left[\frac{p_t^*}{P_s} y_{j,s} - w_s n_{j,s} - \psi M_s h_{j,s}\right] \tag{75}$$

Here $n_{j,t}$ denotes employment at firm j in time t. $h_{j,t}$ denotes hiring and $R_{t,s}^{-1}$ denotes the real interest rate discounted s periods back to time t. This model is solved using Reiter's method in combination with time iteration. This model matches unemployment and interest rate to the same data generated by a parameterized model, only two observed variables are used as the model only had two shocks.

CF. Winberry Approach to Krusell-Smith

Next I will discuss Krusell and Smith (1998) solved with Winberry (2018). The consumer is standard:

$$E[\sum_t = 0^{\infty} \beta^t U(c_t) \tag{76}$$

Utility function is CRRA:

$$U(c-t) = \frac{c^{1-\sigma} - 1}{1-\sigma} \tag{77}$$

The investment equation is:

$$y = c + k' - (1 - \delta)k \tag{78}$$

Agent can be employed or unemployed. If employed works one unit of labor, else doesn't work. This is denoted with an ϵ. Employment status transitions via a Markov matrix.

On the market side, wages and rental rate equal their marginal benefit. In particular if aggregating across all capital ($\bar{k}$) and labor ($\bar{l}$):

$$w = (1 - \alpha)z(\bar{k}/\bar{l})^{\alpha} r = \alpha z(\bar{k}/\bar{l})^{1-\alpha} \tag{79}$$

The states are the distribution of capital and labor, denoted Γ, as well as productivity. For each individual, they optimize via:

$$v(k, \epsilon; \Gamma, z) = max_{c,k'}\{U(c) + \beta E[v(k, \epsilon; \Gamma, z)|z, \epsilon]\} \tag{80}$$

subject to

$$c + k' = rk + wl\epsilon + (1 - \delta)k\Gamma' = H(\Gamma, z, z')k' \geq 0 \tag{81}$$

Each agent optimizes according to their own k and ϵ, however the prices are chosen for them due to the aggregate next period levels of k and ϵ. As such you are solving for a fixed point so that given a distribution of k and ϵ, the next period distribution of k and ϵ also correspond to the prices of k and ϵ that would induce agents to acquire those amounts. Winberry takes a different approach. Since you know the individual's law of motion and you can integrate the distribution to get the aggregate prices that generate that law of motion, you can derive a law of motion for the heterogeneous distribution. If you approximate this distribution via moments, you can derive a law of motion for the moments of the distribution. Then you can apply a non heterogeneous agent approaches to solve this model. In particular, Winberry solves the model in Dynare with

perturbation. For more details on the solution method, see Winberry (2018).

Estimating heterogenous agent models via a likelihood is difficult if one wants to perform full information data with micro-data as well. The reason is that when evaluating the microdata one has to calculate the likelihood of the the micro data marginalizing over the distribution of the state. One does this by sampling from the Kalman filter state space and performing Monte Carlo integration to calculate the probability of the micro data over a large set of samples from the state. Due to the fact that SNPE doesn't require likelihood calculation, this whole process starting from the Kalman filtering and ending up performing likelihood calculations of 100s of thousands points, is avoided.

CG. The Value Function Iteration Bequests Model

Finally, I estimate a model of bequests over time. The model illustrates the benefit of my approach as it cannot be estimated by perturbation because of the kink at the intersection of the consumption versus adjusting illiquid savings intersection. Thus one has to use simulation-based inference to Bayesian estimate this model solved via value function iteration.

This model comes from Jeppe Druedahl's open source model example: `https://github.com/pkofod/vfi/blob/master/Fast%20VFI.ipynb`. The utility function is:

$$u(b_t, c_t) = \frac{[\phi(b_t + \underline{b})^{1-\gamma} + (1-\phi)c_t^{1-\gamma}]^{\frac{1-\rho}{1-\gamma}}}{1-\rho} \tag{82}$$

Here b_t is bequest or non-liquid savings. c_t is consummation. b, ϕ, γ, ρ are all parameters. The bequest utility function (end of life utility) is:

$$\nu(a_t, b_t) = \varphi \frac{(a_t + b_t + \underline{q})^{1-\vartheta}}{1-\vartheta} \tag{83}$$

Here a_t is liquid savings and q, θ, φ are all parameters. The value function is give by:

$$v_t(m_t, n_t, l_t) = \max\{v_t^{keep}(m_t, n_t, l_t), v_t^{adj}(m_t, n_t, l_t)\} \tag{84}$$

$$\text{s.t.} \tag{85}$$

$$x_t = m_t + (1-\tau)n_t \tag{86}$$

Here m_t is beginning of period liquid savings (analogous to end of period a_t) and n_t is beginning of period illiquid savings (analogous to end of period b_t). x_t is the pooled value of total liquid and illiquid savings. l_t is labor income. The post-decision value function is:

$$w_t(a_t, b_t, l_t) = \nu(a_t, b_t), t = T \tag{87}$$

$$w_t(a_t, b_t, l_t) = \mathbb{E}_t\left[\max\{v_t^{keep}(m_{t+1}, n_{t+1}, l_{t+1}), v_t^{adj}(x_{t+1}, l_{t+1})\}\right], t < T \tag{88}$$

$$\text{s.t.} \tag{89}$$

$$l_{t+1} \sim F(l_t) \tag{90}$$

$$m_{t+1} = (1+r_a)a_t + \omega l_{t+1} \tag{91}$$

$$n_{t+1} = (1-\delta)b_t \tag{92}$$

$$x_{t+1} = m_{t+1} + (1-\tau)n_{t+1} \tag{93}$$

Here labor income comes from a distribution $F(l_t)$. Likewise r_a, ω, δ, τ are all parameters. The keep value function is:

$$v_t^{keep}(m_t, n_t, l_t) = \max_{c_t \in [0, m_t]} u(n_t, c_t) + \beta w_t(a_t, b_t, l_t) \tag{94}$$

$$\text{s.t.} \tag{95}$$

$$a_t = m_t - c_t \tag{96}$$

$$b_t = n_t \tag{97}$$

The keep value function is the value function at a point t if the agent chooses to consume and not

withdraw or deposit funds into the illiquid asset, b_t. The adjust value function is:

$$v_t^{adj}(x_t, l_t) = \max_{b_t \in [0, x_t]} v_t^{keep}(m_t, n_t, l_t) \tag{98}$$

$$\text{s.t.} \tag{99}$$

$$m_t = x_t - b_t \tag{100}$$

$$n_t = b_t \tag{101}$$

The adjust value function is the value of adjusting the illiquid asset. Notice the agent is unable to consume this period. This model with it's nonlinear and kinked decision rule cannot be effectively modeled with a method like perturbation. Even so, I use my estimation technique to get a Bayesian posterior distribution. This model matches the liquid asset, illiquid asset and consumption to synthetic data generated by a model.

D Background on Simulation Neural Ratio Estimators

DA. General Approach of Simulation-Based Inference

The differences between the two simulation-based inference approaches roughly centers around what conditional density estimator to use. Sequential Neural Ration Estimators (SNRE) uses a GAN and Sequential Neural Posterior Estimators (SNPE) use a normalizing flow. While one can sample a lot of points from the joint, an iterative approach where one fits the density estimator on sampled data then re-samples leads to lower variance. I will start discussing SNRE, by introducing the GAN

DB. Generative Adversarial Networks

While I focus on SNPE, there are related approaches that use a GAN (Goodfellow et al., 2014) as density estimators, Sequential Neural Ratio Estimation (SNRE). The GAN is an algorithm that uses two competing models moving in tandem to generate data that matches data generated from

the real world. Below is a diagram describing the structure of a basic GAN:

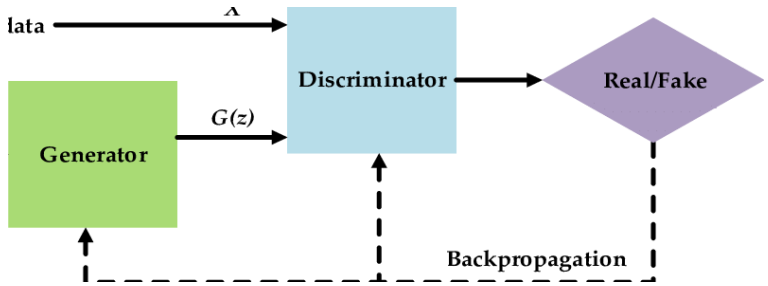

FIGURE XXIII: DIAGRAM OF A GAN

The generator is a neural network (or any differentiable model like a dynamic stuctural model) which attempts to produce data (unconditionally or conditionally via some inputs Z) that resembles data in the real world. The discriminator is another neural network (or any trainable model that can produce a logistic output), which outputs the probability a given input data comes from the underlying data generating process and not from the generator. The two networks play a game where the objective of the generator is to convince the discriminator to assign high likelihood to its output and the objective of the discriminator is to assign the correct probabilistic label, a number between 0 or 1, to input that represents the continuum that the current data point is generated data or real data respectively. One way to think about a GAN is as a simulated method of moments algorithm where the moment conditions automatically evolve over many rounds to pick the most mismatched moments between the true data and the model. When thought in this way, it's intuitive to recognize that the GAN is nearly a full information and efficient estimation procedure. For more thorough proofs see Kaji, Manresa and Pouliot (2020).

Mathematically the discriminator loss is:

$$Loss_D = -E_{x \sim P_{data}(x)}[D(x)] - E_{z \sim P_{sim}(z)}[1 - D(G(z))] \tag{102}$$

This is just a logit loss applied to data either coming from the data generating process or the generator. Likewise the generator's loss is to fool the discriminator:

$$Loss_G = E_{z \sim P_{sim}(z)}[1 - D(G(z))] \tag{103}$$

These two loss functions can be compactly represented as:

$$G, D = \min_G \max_D E_{x \sim P_{data}(x)}[D(x)] + E_{z \sim P_{sim}(z)}[1 - D(G(z))] \tag{104}$$

Equation (104) makes it clear the GAN objective is a minimax game with a solution that is a Nash equlibrium. However what is important for simulation-based estimation is to recover likelihoods/probabilities. The next proposition following Goodfellow et al. (2014) will show how to do that:

Proposition 1.: For a fixed generator, the optimal discriminator will produce a logistic output:

$$D^*(x) = \frac{p_{data}(x)}{p_{data}(x) + p_{generator}(x)} \tag{105}$$

Proof: Using equation (102), for the loss of the generator:

$$Loss_D = \int_x log(D(x))p_{data}(x)dx + \int_z log(1 - D(G(z)))p_{generator}(z)dx \tag{106}$$

$$= \int_x (log(D(x))p_{data}(x) + log(1 - D(x))p_{generator}(x))dx \tag{107}$$

(108)

Any discriminator that maximizes the integral: $\int_x (log(D(x))p_{data}(x) + log(1 - D(x))p_{generator}(x))dx$, maximizes the integral pointwise over all x. Thus the optimal discriminator will set the first order condition of the intergrand to zero:

$$0 = \frac{p_{data}(x)}{D^*(x)} + \frac{p_{generator}(x)}{1 - D^*(x)} \tag{109}$$

Rearranging equation (109) gets the result of the proposition. ■

Then one can use this proposition along with the discriminator output to estimate a likelihood ratio.

DC. Normalizing Flows

In this section, I will discuss how a particular deep learning model, the neural autoregressive flow (Huang et al., 2018), is structured. Then I will discuss how to use the change of variables formula for a random variable to derive a likelihood of a sample under the flow. I will also attempt to highlight advantages flows have over KDEs that make the SNPE algorithm more powerful that the traditional Kristensen and Shin (2012) approach, including the ability to do conditional density estimation, high dimensional density estimation, and the ability to sample and maximum likelihood estimate without resort to rejection sampling.

Take a function f_i which turns a random variable z_i into the random variable z_{i+1}. The formula for $q(z_{i+1} = f_i(z_i))$ given $p(z_i)$ is the familiar change of variables:

$$q(z_{i+1} = f_i(z_i)) = p(z_i = f_i^{-1}(z_{i+1})|det\frac{df_i^{-1}(z_{i+1})}{dz_{i+1}})| = p(z_i)|det\frac{df_i(z_i)}{dz_i}|^{-1} \tag{110}$$

Given a series of transformations $\theta = f_K \circ f_{K-1} \circ ...f_1(z_1)$, then the formula for the whole sequence of change of variables is:

$$ln(q(\theta)) = ln(p(z_1)) - \sum_{k=0}^{k=K} |det\frac{df_k(z_k)}{dz_k}| \tag{111}$$

Each f_i is called a bijector and a chain of f's in conjunction with the initial base z_1 is called a flow. Given this information, we can create a flexible function that can convert a simple distribution–for

example a multivariate normal with mean vector $\bar{\mathbf{0}}$ and standard deviation vector $\bar{\mathbf{1}}$–into many different "non-parametric" distributions. In fact Huang et al. (2018)–the model this paper uses–shows that by chaining together f's, under some assumptions, a flow can approximate any continuous distribution from a normal distribution.

Before moving on, discussing some syntax: a subscript under a variable refers to the layer in the flow, ie the k that corresponds to which f_k. As all these elements are vectors and f is a vector valued function, a supercript, j, indicates the element j of the vector z_k. Since I want to use a flow to perform density estimation, here are some useful properties. First if I know θ in order to calculate the likelihood of theta, I need to go from z_K to z_1 as that value is required in the change of variable equation. However, if I want to be able to sample from the flow, I need to be able to sample z_1, then feed it into the flow to get θ. Thus implies the f_k's must be bijections, which is why I call them bijectors.

A simple but inflexible function that is a bijection is an affine transformation:

$$z_{k+1} = a_k * z_k + b_k \tag{112}$$

and it's inverse:

$$z_k = \frac{z_{k+1} - b_k}{a_k} \tag{113}$$

In order to make the bijector more expressive, I will go elementwise through the vector z_k and condition later elements on the earlier ones.

$$z^0_{k+1} = a^0_k * z^0_k + b^0_k z^{j>0}_{k+1} = a^j_k(z^{j-1}_k ... z^0_k) * z^j_k + b^j_k(z^{j-1}_k ... z^0_k) \tag{114}$$

Here the a^j_k and b^j_k terms are functions, that take in the lagged z terms and output coefficients. In this way the bijector maintains its ability to remain invertable, while also having flexibility to

approximate arbitrary monotonic functions. If you know the vector z_k, you can easily calculate z_{k+1}. If you know z_{k+1} you can initially calculate z_k^0 and knowing that you can calculate z_k^1 and via induction you can calculate the rest of the z_k's. Additionally permutation of the elements of z_k is also a bijection and this allows one to model arbitrary conditioning relationships between dimensions. Finally, in order to increase flexibility one can add a bijective nonlinearity like a ReLU:

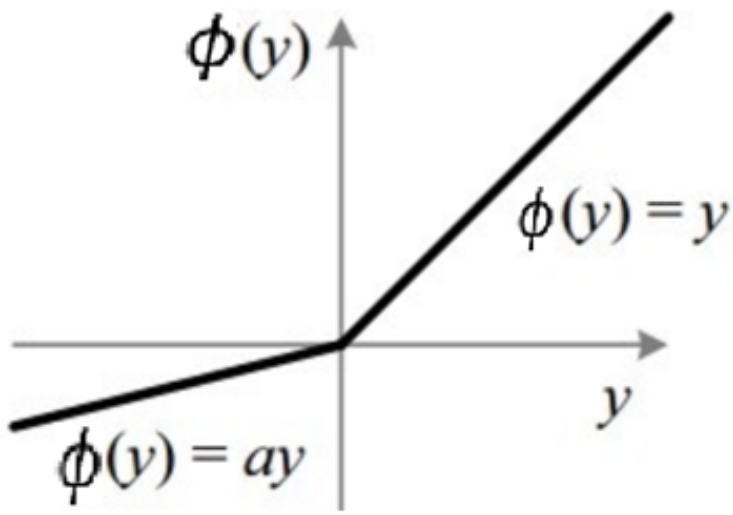

Figure XXIV: Diagram of a ReLU non-linearity

Huang et al. (2018) show that a flow comprised of bijectors with logit and inverse logit nonlinearities, can map any continuous distribution to any other continuous distribution. As a result, flows in this class, generally have "non-parameteric" like properties that allow it to map any distribution converge in distribution to any other continuous distribution. Likewise the "non-parameteric" nature of flows, GANs and mixture of Gaussians, allow the SNPE and related algorithms to converge under the KL divergence to the Bayesian posterior (Papamakarios and Murray, 2016), (Greenberg, Nonnenmacher and Macke, 2019).

DD. Sequential Neural Ratio Estimation

In SNRE, Durkan, Murray and Papamakarios (2020) use a GAN, not to match a underlying data like in proposition 1, but to contrast between two different distributions to estimate a density. In macro, the dataset is typically a single data point with correlated timesteps. Using a GAN to discriminate between real data and fake data would not work because the GAN would learn a delta function for the real data. In order to ameliorate this, the authors use contrastive learning where they differentiate between the generated distribution and the independent distribution, $x, \theta \sim p(x)p(\theta)$, where x is sampled uniformly from the empirical distribution of simulated data. Then they contrast the independent distribution, with the true joint: $x, \theta \sim p(x, \theta) = p(x|\theta)p(\theta)$. Furthermore, knowing the optimal discriminator gives the ratio discussed in (105). The authors then recover the ratio: $r(x, \theta) = \dfrac{p(x, \theta)}{p(x)p(\theta)}$, using the fact that $\dfrac{p(x, \theta)}{p(x, \theta) + p(x)p(\theta)}$ equals $\dfrac{r}{r+1}$. Thus, the density of the posterior $p(\theta|x) = r(x, \theta)p(\theta)$, where $p(\theta)$ is the prior. This extends Kaji, Manresa and Pouliot (2020) GAN approach in allowing estimation of models with a latent time dimension. One can extend this to multi-round inference where one uses the GAN as a better proposal distribution than the prior, by sampling θ from the GAN using MCMC. Then one uses importance sampling to deal with the mismatch in proposal distribution and prior. See Durkan, Murray and Papamakarios (2020), Hermans, Begy and Louppe (2020) and the SNRE theoretical sections of the appendix for more information.

Algorithm 2: SNRE Algorithm

Input: Simulator $p(x|\theta)$, prior $p(\theta)$, data x_0, discriminator $d_\phi(x,\theta)$, Rounds R, Samples S;

Initialize: Posterior $p^{(0)} = p(\theta)$, data set D = {};

for $i \leftarrow 1$ **to** R **do**

  Sample $\theta^{(n)} \sim p^{(i-1)}$ for $n = 1...S$ with MCMC;

  Simulate $x^{(n)} \sim P(x|\theta^{(n)})$ for $n = 1...S$;

  Concatenate data $D = D \cup \{x^{(n)}, \theta^{(n)}\}_{n=1}^{S}$;

  while $d_\phi(x,\theta)$ *not converged* **do**

    Sample $\{x^{(i)}, \theta^{(i)}\}_i^B \sim D$ from D;

    Sample K contrasting samples for each i sample

    $\{x'^{(k)}\}_k^{B*K}, \{\theta'^{(k)}\}_k^{B*K} \sim D\{x\}, D\{\theta\}$, breaking the joint distribution of D into independent marginals;

    Train the GAN using the GAN loss function from the two distributions x, θ and x', θ' and a softmax/multinomial logit objective $\sum_i^B log(\frac{exp(d_\phi(x^{(i)}, \theta^{(i)})}{\sum_k^K exp(d_\phi(x^{(k)}, \theta^{(k)})})$;

  end

  Update posterior $p^{(i)} \propto exp(d_\phi(x,\theta))p(\theta)$;

end

DE. Sequential Neural Variational Inference (SNVI)

In SNVI, Glöckler, Deistler and Macke (n.d.), one performs SNRE, but instead of MH-MCMC sampling from the GAN posterior, one projects the distrbiution derived by the GAN unto a normalizing flow with variational inference. As discussed in this article, MH-MCMC takes a long time to converge to the posterior. Since one can calculate the pdfs of a sample point using the GAN ratio estimation, one can sample from a normalizing flow, calculate the pdf of the sample under both the flow and the GAN, and use this to minimize the KL divergence between the flow and the GAN. Thus the flow inherits the distribution of the GAN, and one can sample directly from the flow without resorting to MH-MCMC

Algorithm 3: SNVI Algorithm

Input: Simulator $p(x|\theta)$, prior $p(\theta)$, data x_0, discriminator $d_\phi(x,\theta)$, Rounds R, Samples S;

Initialize: Posterior $p^{(0)} = p(\theta)$, data set D = {};

for $i \leftarrow 1$ **to** R **do**

  Sample $\theta^{(n)} \sim f_{\phi'}^{(i-1)}(\theta)$ for $n = 1...S$;

  Simulate $x^{(n)} \sim P(x|\theta^{(n)})$ for $n = 1...S$;

  Concatenate data $D = D \cup \{x^{(n)}, \theta^{(n)}\}_{n=1}^{S}$;

  while $d_\phi(x,\theta)$ *not converged* **do**

    Sample $\{x^{(i)}, \theta^{(i)}\}_i^B \sim D$ from D;

    Sample K contrasting samples for each i sample

    $\{x'^{(k)}\}_k^{B*K}, \{\theta'^{(k)}\}_k^{B*K} \sim D\{x\}, D\{\theta\}$, breaking the joint distribution of D into independent marginals;

    Train the GAN using the GAN loss function from the two distributions x, θ and x', θ' and a softmax/multinomial logit objective $\sum_i^B log(\frac{exp(d_\phi(x^{(i)}, \theta^{(i)})}{\sum_k^K exp(d_\phi(x^{(k)}, \theta^{(k)})})$;

  end

  Train normalizing flow with variational inference

  $f_{\phi'}^i = argmin_{\phi'}(KL(f_{\phi'}^i(\theta)||exp(d_\phi(x,\theta))p(\theta))$;

end

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